

Efficiency and Equity Through Intertemporal Coordination in Groups^{*}

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Abstract

This paper experimentally examines how group size and composition affect efficiency and equity in intertemporal coordination. In a modified market entry game, players repeatedly choose between two capacity-constrained markets, one yielding a higher payoff than the other. Efficiency and equity can be achieved through intertemporal coordination, with players taking turns to enter the high-payoff market. We find that smaller groups readily establish turn-taking, whereas larger groups often fail, exhibiting less alternation and less stable coordination. Effects of group identity composition depend on group size. In smaller groups, heterogeneous identity is associated with less efficient coordination early on, but performance converges quickly. In larger groups, homogeneous identity is associated with increasingly unequal outcomes, alongside emergent patterns of exploitation and accommodation. Overall, shared identity does not uniformly improve outcomes, highlighting how coordination complexity and identity composition jointly shape the emergence and persistence of behavioral patterns in intertemporal cooperation.

JEL codes: C73, C91, C92, D91

Keywords: Intertemporal coordination, market entry game, group size, identity, experiment

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1 Introduction

Competition over limited resources often creates a tension between equity and efficiency. In such settings, intertemporal coordination, or taking turns, can offer a solution. However, successful turn-taking requires more than goodwill; it depends on patience and trust that delaying immediate gains will be reciprocated in the future. While such arrangements can sustain fair and efficient outcomes, they are often difficult to establish and maintain, especially when short-term gains are uneven.

What explains these failures, and what allows some groups to succeed? A key determinant is the social environment in which coordination occurs. Individuals' willingness to wait their turn, and their belief that others will do the same, depend on who they interact with, how cohesive the group is, and how difficult coordination is to sustain. Social structures that foster cohesion and trust may support restraint, whereas those that increase strategic uncertainty or social distance may undermine it. This raises a central question: how do features of the social environment shape the emergence and persistence of intertemporal cooperation?

In this paper, we focus on two key aspects of social structure: group size and group composition. Group size affects coordination complexity: as groups grow, coordination becomes more demanding both cognitively and strategically (Dunbar, 1992). Group composition, by determining whether members share a common identity, may shape social preferences, interpretations of others' actions, and social norms (Chen et al., 2014; Charness et al., 2007; Chuah et al., 2016). These dimensions may also interact. While shared identity can promote cohesion and cooperation, distinct identities may provide coordination cues in more complex environments. The effect of group composition may therefore vary with group size, raising the question of whether cohesion or coordination cues are more important.

To study this, we conduct a laboratory experiment in which groups play a repeated market-entry coordination game building on Camerer and Lovo (1999). In each round, participants choose between two markets with limited capacity, one of which yields a higher payoff. While efficiency can be achieved if half the group selects each market, equity requires members to rotate access to the high-payoff market over time. We implement a 2×3 between-subjects design, varying group size (two-player vs. four-player) and group composition (homogeneous vs. heterogeneous with respect to induced social identities, or with no identity attributes).

Our results show that group size is the dominant source of variation in intertemporal cooperation. Two-player groups almost always establish turn-taking, achieving both high efficiency and equity, whereas approximately 74 percent of four-player groups fail to sustain such arrangements.

These results highlight that even modest increases in group size substantially hinder intertemporal coordination. The effects of group composition are more nuanced and depend on group size. In two-player groups, heterogeneous identity initially slows the emergence of turn-taking, but this effect diminishes over time. In four-player groups, identity primarily affects equity rather than efficiency: while inefficiencies persist across all group compositions, homogeneous groups become significantly more unequal over time.

Coordination success in Round 1 plays a limited role in explaining these differences. Instead, the evidence points to the persistence of behavioral patterns that emerge during early play. In two-player groups, turn-taking quickly becomes the dominant pattern of behavior. In contrast, four-player groups exhibit greater behavioral heterogeneity early on, with multiple patterns of play coexisting and outcomes varying more widely across groups. These behavioral patterns are highly persistent, and behavioral composition during early play predicts subsequent group performance even after accounting for efficiency and equity over the same period. Notably, contrary to standard predictions, shared identity can sustain inequitable outcomes, with some members persistently claiming the high-payoff market while others accommodate.

Taken together, our findings suggest that coordination complexity, associated with group size, plays a central role in whether stable turn-taking can emerge, whereas group composition shapes the form of coordination failures. In particular, shared identity does not necessarily improve outcomes: in more complex environments, it may instead sustain inequitable patterns of exploitation and accommodation. To our knowledge, this is the first study to provide clean evidence that even a modest increase from two to four players can generate large and persistent failures in intertemporal coordination, and that these failures operate through the composition of behavioral types rather than through early path dependence.

More broadly, our results show that social structures shape not only whether coordination is achieved, but also how outcomes are distributed. These insights are relevant for many real-world settings such as economic policy, climate agreements, and the governance of common resources, where groups repeatedly allocate scarce resources over time. In increasingly diverse and polarized societies, understanding how social structure shapes intertemporal coordination is essential for organizations, communities, and policymakers seeking to design institutions that balance efficiency and equity.

Section 2 reviews the related literature and positions our contribution. Section 3 describes the experimental design, and Section 4 presents the research questions. Section 5 reports the results, and Section 6 concludes.

2 Related literature

Our study relates to a broad literature on cooperation and coordination in groups, spanning foundational models of collective action (Olson, 1971; Hardin, 1968; Axelrod, 1984), experimental evidence on cooperation (Fehr and Gächter, 2000; Ledyard, 1995), and institutional approaches to sustaining it (Ostrom, 1990). We focus on how *social structures* shape intertemporal coordination, in particular the interaction between group size and social identity. Table 1 summarizes representative studies along these dimensions and highlights how our setting differs.

Table 1: Summary of Related Literature

	Game Design			Research Question		
	Game (Repeated/Static)	Group Sizes	Intertemporal Eff. & Eq. †	Group Size	Social Identity	Turn Taking
<i>Panel A: Experiments on Group Size and Coordination</i>						
Van Huyck et al. (1990)	Minimum Effort Game (R)	2, 14, 15, 16	×	✓	×	×
Knez and Camerer (1994)	Minimum Effort Game (R)	3, 6	×	✓	×	×
Weber (2006)	Minimum Effort Game (R)	2 to 12	×	✓	×	×
Huck et al. (2004)	Cournot Oligopoly (R)	2 to 5	×	✓	×	×
Nosenzo et al. (2015)	PGG (R)	2, 3, 4, 8	×	✓	×	×
Isaac et al. (1994)	PGG (R)	4, 10, 40, 100	×	✓	×	×
Weimann et al. (2019)	PGG (R)	8, 20, 30, 40, 60, 100	×	✓	×	×
Barcelo and Capraro (2015)	PGG & PD (S)	4, 40 / 2, 11	×	✓	×	×
<i>Panel B: Experiments on Identity and Coordination</i>						
Chen and Chen (2011)	Minimum Effort Game (R)	2	×	×	✓	×
Chen et al. (2014)	Minimum Effort Game & PD (S)	2	×	×	✓	×
Eckel and Grossman (2005)	PGG (R)	5	×	×	✓	×
Goette et al. (2006)	PD (S)	2	×	×	✓	×
Charness et al. (2007)	BoS & PD (S)	2	×	×	✓	×
Attanasi et al. (2016)	BoS & Entrance Game (S)	2	×	×	✓	×
<i>Panel C: Experiments on Intertemporal Coordination</i>						
Kaplan and Ruffle (2012)	Entry Game (R)	2	✓	×	×	△
Bruttel and Güth (2018)	Sequential Best Shot (R)	2	✓	×	×	△
Kuzmics et al. (2014)	BoS (R)	2	✓	×	×	△
Agranov et al. (2024)	BoS (R)	2	✓	×	×	✓
Bjedov et al. (2016)	Coordination Game (R)	2	✓	×	×	✓
Cason et al. (2013)	Common Pool Resources (R)	2	✓	×	×	✓
Riyanto and Roy (2019)	Threshold Public Goods (R)	3	✓	×	×	✓
Zillante (2017)	Market Entry Game (R)	2, 3, 4	✓	✓	×	✓
This Study	Market Entry Game (R)	2, 4	✓	✓	✓	✓

PGG = Public Goods Game; PD = Prisoner’s Dilemma; BoS = Battle of the Sexes. R = Repeated; S = Static.

† Indicates whether the design allows for intertemporal coordination of efficiency and equity.

△ Turn-taking is observed but not the primary focus.

In studies on group size (see Panel A), effects depend on the coordination environment. In Cournot oligopoly, minimum effort, and prisoner’s dilemma games, larger groups tend to exhibit higher coordination failure (Van Huyck et al., 1990; Huck et al., 2004; Barcelo and Capraro, 2015). However, this pattern is not universal: gradual increases in group size can mitigate the

breakdown in minimum effort games (Weber, 2006). In public goods games, evidence is mixed and depends on the marginal per capita return (Nosenzo et al., 2015). This heterogeneity suggests that group size effects are context-dependent, motivating our focus on intertemporal coordination, where coordinating behavior over time adds further complexity.

A parallel literature, building on Tajfel et al. (1971) and Tajfel and Turner (1979), examines how social identity shapes coordination (see Panel B). Experimental evidence is largely consistent: across a variety of games, shared identity improves coordination and cooperation (Eckel and Grossman, 2005; Chen et al., 2014; Attanasi et al., 2016), consistent with the well-established effects of identity on social preferences (Li, 2020). Nevertheless, it remains unclear whether identity can also facilitate coordination in intertemporal environments by providing cues under higher strategic uncertainty (Smaldino, 2019).

Most existing studies examine settings in which efficiency and equity can be achieved within a single period. They therefore do not address intertemporal coordination, where these outcomes must be sustained over time. Panel C summarizes the relatively limited experimental evidence in such environments.¹ Existing studies suggest that turn-taking can emerge, primarily in two-player groups. Several papers document the spontaneous adoption of alternating strategies in repeated interactions (Kaplan and Ruffle, 2012; Bruttel and Güth, 2018; Kuzmics et al., 2014), and others further show that communication, teaching or similar levels of patience can facilitate the emergence of turn-taking (Bjedov et al., 2016; Cason et al., 2013; Agranov et al., 2024).

Evidence in larger groups remains limited: Riyanto and Roy (2019) show that turn-taking can emerge in three-player groups under homogeneous endowments but breaks down under heterogeneity. A closely related (though unpublished) study, Zillante (2017), examines whether groups of different sizes can coordinate on turn-taking using a multiperiod signaling device. They find that turn-taking emerges in small groups but becomes substantially more difficult as group size increases, with signaling playing a greater role in larger groups.²

Our paper instead examines how group size and social identity jointly shape the emergence and stability of turn-taking. We show that they affect not only whether intertemporal coordination arises, but also the behavioral patterns that emerge, highlighting interactions between coordination complexity and social structure absent from existing work. To our knowledge, this is the first study to jointly examine group size, social identity, and intertemporal coordination.

¹In environments with repeated interaction and asymmetric payoffs, achieving both efficiency and equity requires intertemporal coordination.

²While Zillante (2017) also study a market entry game, their payoffs are derived from a k -player Cournot model with a fixed outside option. As a result, average earnings in the efficient outcome vary with group size, making outcomes less directly comparable across group sizes than in our setting.

3 Methodology

3.1 Experiment Design

Participants play a market entry game adapted from [Camerer and Lovallo \(1999\)](#) in fixed groups under a 2×3 between-subjects design. In the first treatment dimension, we vary the group size. Groups are either *two-player* (2P) or *four-player* (4P). In the second treatment dimension, we vary the group composition. Groups are either *homogeneous* (all members have the same social identity), *heterogeneous* (half the members belong to each of two social identities), or have *no identity*. All groups are symmetric by construction, abstracting from any imbalance across social identities (if any). Within groups, each subject chooses between two markets, A or B; payoffs depend on i) the market chosen and ii) how many group members choose each market. The experiment consists of two main stages.

Stage 1: Identity Inducement

In the first stage, for treatments that involve social identities, we induce identities in participants by adopting the pseudo-minimal group paradigm as in [Chen and Li \(2009\)](#). Participants are randomly assigned into two social identity categories of equal size: *red* or *blue*. Subsequently, to build up their identities, members of each social identity complete a joint task which involves spotting the differences between pairs of pictures. For each pair, members can discuss solutions via the built-in chat function, and a selected representative will submit their answers. This procedure ensures equal earnings for all members within an identity category, aligning incentives and encouraging stronger mutual identification. Across the two picture pairs, there were a total of 11 differences; participants earned 200 ECUs for each correctly identified difference. At the end of the joint task, they learn about their earnings and proceed to the second stage.³

In treatments without social identities, participants are *not* assigned to *blue* or *red* social identities. Instead, they complete the same joint task with a randomly selected set of unlabeled participants.

³There is evidence that our identity inducement was successful. Using the Inclusion of Others in Self (IOS) scale to measure closeness to different identity classes, we find that individuals, on average, feel 1.40 points closer to ingroup members than outgroup members, $p < 0.001$. A third-party allocation task at the end of the experiment also shows that they distribute 308.59 more ECUs to members of the ingroup than to members of the outgroup, $p < 0.001$. This is consistent with the broader evidence that the minimal group paradigm reliably produces in-group favoritism in laboratory settings ([Lane, 2016](#)).

Stage 2: Market Entry Game

In the second stage, participants (now induced with a *blue* or *red* identity) are matched into groups to play a 60-round market-entry game. Group size and identity composition are set by treatment: two-player groups have 2 members and four-player groups have 4 members. Within each group, participants are randomly assigned an ID that remains fixed throughout: 1–2 in two-player groups or 1–4 in four-player groups. Participants are fully informed about their matching and IDs, which remain fixed throughout the market entry game.⁴

Participants choose between Market A and Market B in each round. Each market has a capacity equal to half the group size. Choosing A yields 400 ECUs and choosing B yields 200 ECUs, provided the number of choosers does not exceed a market’s capacity (see Table 2). In the one-shot game, the efficient pure-strategy Nash equilibrium has half the group choosing A, and the other half B. Over repeated rounds, achieving both efficiency and equity therefore requires turn-taking: members alternate between A and B such that capacities are respected. With efficient turn-taking, average earnings are 300 ECUs per round, regardless of group size.

Participants are able to observe the full history of choices made by everyone in their own group. Their earnings in the market entry game are the sum of payoffs across all 60 rounds, making turn-taking over time payoff-relevant.

Table 2: Payoff Summary

Number who choose Market A	0	1	2	3	4
<i>Two-player group</i>					
Payoff if in Market A	-	400	50	-	-
Payoff if in Market B	50	200	-	-	-
<i>Four-player group</i>					
Payoff if in Market A	-	400	400	50	50
Payoff if in Market B	50	50	200	200	-

Number who choose Market B is (i) 2 minus number who choose Market A in two-player groups and (ii) 4 minus number who choose Market A in four-player groups.

Procedures

Participants were recruited via mass emails at Nanyang Technological University from August to September 2023; they were largely undergraduate students from a variety of majors. In total,

⁴Note that while we refer to these as “groups” for exposition, the term was not used during the experiment to preserve neutral framing. The choice of markets was also framed as a choice between Activity A or Activity B. See Appendix F for more details.

there were 418 participants across 18 different sessions with approximately 22 to 25 groups in each treatment, see Table 3. Each session was randomly assigned to an identity-inducement condition (identity inducement/no identity inducement) and a group size (two-player/four-player). In sessions with identity inducement, participants were further randomized into homogeneous or heterogeneous groups. All sessions were programmed in z-Tree (Fischbacher, 2007). The study obtained ethical approval from Nanyang Technological University (IRB-2022-611). See also Appendix F for the full set of instructions and screenshots of the study.⁵

Table 3: Treatment Descriptions

	No Identity (No ID)	Homogeneous (Hom)	Heterogeneous (Het)
Two-player (2P)	$n = 50$ (25 groups)	$n = 46$ (23 groups)	$n = 46$ (23 groups)
Four-player (4P)	$n = 92$ (23 groups)	$n = 88$ (22 groups)	$n = 96$ (24 groups)

At the beginning of each session, detailed printed instructions of Stage 1 were provided to participants and read aloud to them. There was thus common information, which has been shown to be important for the identity inducement procedure (Chen et al., 2021). This was followed by Stage 1 of the experiment. After Stage 1, detailed printed instructions of Stage 2 were then provided and also read aloud to participants. This was followed by a quiz to test their understanding of the market entry game before beginning the first round of Stage 2.⁶ All questions needed to be answered correctly before they could proceed. At the end of Stage 2, we elicited participants' risk preferences and in-group preferences via incentivized tasks. This was followed by an online post-experiment survey covering questions on demographics, behavior and perceptions about their groups and matches. A summary of their payoffs was then shown. The exchange rate was set at 1000 ECUs for SGD 1. Each session lasted around 1 hour, and participants earned an average of 20.21 SGD (approximately USD 15.67), including a show-up fee of SGD 2.

⁵The study was not pre-registered; the research questions and hypothesized mechanisms were developed in the grant proposal that funded the project.

⁶In the post-experiment survey, participants also indicated that they understood the instructions well. On a Likert scale from 0 (Do not understand at all) to 6 (Understand completely), the mean was 5.37 on average, with a standard deviation of 0.94.

3.2 Analysis Approach

For our analysis, we focus on two main group-level outcomes: efficiency and equity. Both of these outcomes are important because successful intertemporal coordination via turn-taking can be characterized by both high efficiency and high equity.

Efficiency

For group g and any even window length T (in rounds), define efficiency as

$$Eff_T^g = \frac{\bar{\pi}_T^g - \bar{\Pi}_T^{\min}}{\bar{\Pi}_T^{\max} - \bar{\Pi}_T^{\min}} \in [0, 1].$$

Here, $\bar{\pi}_T^g$ is the group's *per-person, per-round* average payoff over the T -round window, and $\bar{\Pi}_T^{\min}$ and $\bar{\Pi}_T^{\max}$ are the *theoretical* lower and upper bounds on such averages (given the game's rules). Under our parameters, $\bar{\Pi}_T^{\min} = 50$ and $\bar{\Pi}_T^{\max} = 300$. Thus, $Eff_T^g = 0$ when $\bar{\pi}_T^g = 50$ and $Eff_T^g = 1$ when $\bar{\pi}_T^g = 300$.

Equity

For group g and any even window length T (in rounds), define the group's *inequity* as

$$\Delta_T^g = \max_{i \in g} \bar{\pi}_{i,T}^g - \min_{i \in g} \bar{\pi}_{i,T}^g,$$

where $\bar{\pi}_{i,T}^g$ is member i 's *per-person, per-round* average payoff in group g over the T -round window. Let $\Delta_T^{\min}$ and $\Delta_T^{\max}$ denote the *theoretical* lower and upper bounds on inequity under the game's rules ($\Delta_T^{\min} = 0$ and, under our parameters, $\Delta_T^{\max} = 350$). We measure *equity* by normalizing inequity to $[0, 1]$ and reversing its scale:

$$Equity_T^g = 1 - \frac{\Delta_T^g - \Delta_T^{\min}}{\Delta_T^{\max} - \Delta_T^{\min}} \in [0, 1].$$

Thus, $Equity_T^g = 1$ when all members earn the same ($\Delta_T^g = 0$), and $Equity_T^g = 0$ when inequity is maximal ($\Delta_T^g = \Delta_T^{\max} = 350$).

In practice, the extreme cases of zero efficiency or zero equity are unlikely, but we normalize both measures to allow for these possibilities.⁷ The stage game admits multiple Nash equilibria

⁷Minimum efficiency arises under complete miscoordination when all members choose the same market, yielding

(NE). Consequently, there are also numerous sub-game perfect Nash equilibria (SPNE).⁸ We focus on two specific SPNE which we use as benchmarks in Section 5. The first is a *turn-taking equilibrium*: in every round exactly half the group chooses Market A, with players alternating between A and B across rounds, giving $(Eff_T^g, Equity_T^g) = (1, 1)$. The second is a *no-turn-taking equilibrium*: the same half always chooses Market A, which results in efficiency but inequity, with $(Eff_T^g, Equity_T^g) = (1, 0.43)$.

4 Research questions

In the repeated market entry game, we have seen that there are multiple SPNE that differ in their efficiency and equity. Given that people often value both efficiency (Charness and Rabin, 2002) and equity (Rabin, 1993; Fehr and Schmidt, 1999), we are interested in how well groups are able to achieve the dual objectives of both efficiency and equity through turn-taking arrangements (in other words, intertemporal coordination). We are particularly interested in how different social environments influence their ability to reach such agreements.

Two main factors are likely to drive their ability to reach a turn-taking arrangement. First, the difficulty of coordination in general: if the decision environment is strategically and cognitively complex, coordinating on any equilibrium including the turn-taking one will be harder. Second, the desire of group members to reach a turn-taking agreement: a stronger desire will make it more likely that they do so. We focus on three main research questions.

1) How does group size influence the ability to coordinate intertemporally?

Our first research question concerns the effects of group size. Given the greater complexity of intertemporal coordination, we expect larger groups to perform worse. A larger group size increases the strategic complexity of the environment and may also amplify free-riding incentives, both of which discourage turn-taking.⁹ This is analogous to the minimum effort game, where larger groups exhibit poorer coordination due to increased strategic uncertainty (Van Huyck et al., 1990).

$(Eff_T^g, Equity_T^g) = (0, 1)$. Minimum equity arises (in four-player groups) when one member chooses Market A and all others choose Market B, yielding $(Eff_T^g, Equity_T^g) = (0.35, 0)$.

⁸Any strategy profile where half the group chooses Market A is an NE. Any sequence of NE strategy profiles in each round is hence an SPNE. See also Appendix E for further theoretical discussion.

⁹Theoretically, following Lau and Mui (2008, 2012), who analyze turn-taking with independent randomization strategies, one can show that as group size increases, the probability of reaching a turn-taking SPNE after phases of mixed-strategy play declines; see Appendix E.

2) How does group identity affect the ability to coordinate intertemporally?

Our second research question explores the effects of group composition. The impact is ambiguous: shared identity can enhance coordination by fostering closeness (Chen and Chen, 2011) and consequently desires to take turns, but may hinder it by removing useful identity-based labels (Smaldino, 2019). In both no identity and homogeneous groups, coordination differences likely stem only from varying degrees of closeness, as identity labels are absent. In contrast, heterogeneous groups introduce identity labels (*blue/red*) that may affect coordination, particularly in large groups. In smaller (two-player) groups, the value of labels might be limited, as the *me/other* distinction functions similarly to identity labels. In larger (four-player) heterogeneous groups, however, identity labels may act as coordination tools.¹⁰ To examine these dynamics, we focus on comparisons of *homogeneous* vs. *no identity* groups, and *homogeneous* vs. *heterogeneous* groups across both two-player and four-player groups.

3) What explains the differences in intertemporal coordination across groups varying in size and identity composition?

Our third research question examines the mechanisms by which intertemporal coordination arises. To do so, we explore the patterns of strategies played by participants in the market entry game and how they relate to previously observed outcomes.

5 Results

5.1 Preliminaries

Before addressing our questions, we first discuss general behavior in the market entry game. Table 4 presents some descriptive statistics. On average, participants chose Market A 33.81 times out of 60 rounds (56% of the time). In round 1, there was generally miscoordination in choices, with only 44% of groups having exactly half their members in each market. This however improves over time, and across all 60 rounds, groups were on average able to coordinate in 44 rounds (74% of the time).¹¹ Consistent with taking turns, a majority (96%) also switched

¹⁰Theoretically, it can be shown that in large heterogeneous groups, there exists an SPNE where coordinating on taking turns based on identity labels after rounds of mixed strategy play can be more successful than an identity independent mechanism, see Appendix E.

¹¹These overall rates of coordination are significantly higher than if participants chose randomly. If participants chose Market A and Market B at random, then they would coordinate 50% of the time in two-player groups and 37.5% of the time in four-player groups.

markets at least once, with participants switching 33 times on average. Nevertheless, there is no perfect intertemporal coordination, with efficiency and equity in groups at 0.77 and 0.90 on average. On average, participants earned 231 ECUs in each round.

Table 4: Descriptive Statistics

	2P			4P			Overall
	Hom	Het	No ID	Hom	Het	No ID	
Times chose Market A	31.15 (9.36)	32.33 (8.16)	31.46 (5.57)	35.47 (14.40)	34.74 (11.55)	34.60 (13.04)	33.81 (11.56)
Coordinated in Round 1	0.61 (0.50)	0.48 (0.51)	0.52 (0.51)	0.32 (0.48)	0.46 (0.51)	0.22 (0.42)	0.44 (0.50)
Times coordinated	56.13 (9.24)	52.39 (8.64)	55.48 (7.73)	32.91 (11.48)	32.04 (11.30)	35.74 (12.84)	44.27 (14.76)
Times switched	50.85 (17.56)	47.33 (18.71)	50.20 (16.99)	23.31 (18.90)	25.59 (18.33)	26.43 (19.93)	33.41 (21.86)
Group efficiency	0.94 (0.15)	0.87 (0.14)	0.92 (0.13)	0.62 (0.16)	0.61 (0.16)	0.66 (0.18)	0.77 (0.21)
Group equity	0.94 (0.16)	0.95 (0.12)	0.97 (0.08)	0.82 (0.13)	0.87 (0.07)	0.83 (0.15)	0.90 (0.13)
Round earnings (ECUs)	283.88 (47.65)	268.30 (42.48)	281.17 (35.21)	204.50 (51.46)	202.51 (44.98)	215.24 (54.51)	231.33 (58.32)

Means with standard deviation in parentheses. Group efficiency and equity are calculated for $T = 60$.

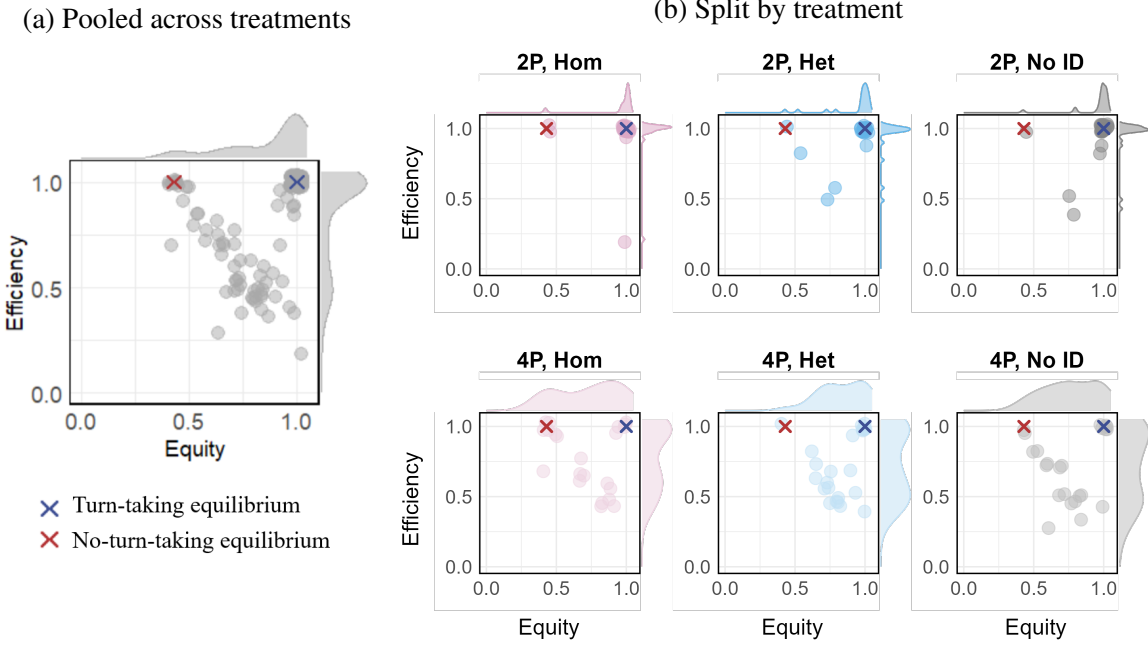
Table 4 also shows that despite relatively similar rates of choosing Market A across the six treatments, there are large and significant differences in efficiency and equity across them (Kruskal-Wallis, $p < 0.001$ for each). Overall, the dominant source of variation appears to be group size rather than identity composition. Efficiency and equity both differ significantly across two-player and four-player groups (Mann-Whitney, $p < 0.001$ for each). By contrast, differences across identity compositions are modest (Kruskal-Wallis: $p = 0.180$ for efficiency; $p = 0.872$ for equity).

Figure 1 examines groups' efficiency and equity over the final 20 rounds relative to our two benchmarks – after they have had time for convergence to equilibrium. A majority of two-player groups achieve the turn-taking equilibrium benchmark with high efficiency and high equity. In four-player groups, there is a wider range of outcomes. While some groups achieve the turn-taking equilibrium benchmark, others exhibit an efficiency-equity tradeoff, with some at the no-turn-taking equilibrium benchmark with high efficiency but low equity.

In subsequent sections, we examine in greater detail how the group-level outcomes of efficiency and equity differ across treatments. To account for the possibility that the dynamics of turn-taking might have multiple rhythms, we present all measurements for the windows $T = 4$

and $T = 20$ to identify both short-term and long-term alternating strategies.¹² We focus our quantitative analysis on the window of $T = 20$; results are qualitatively similar when using the window of $T = 4$, see Appendix B.

Figure 1: Group Efficiency and Equity (Last 20 Rounds)



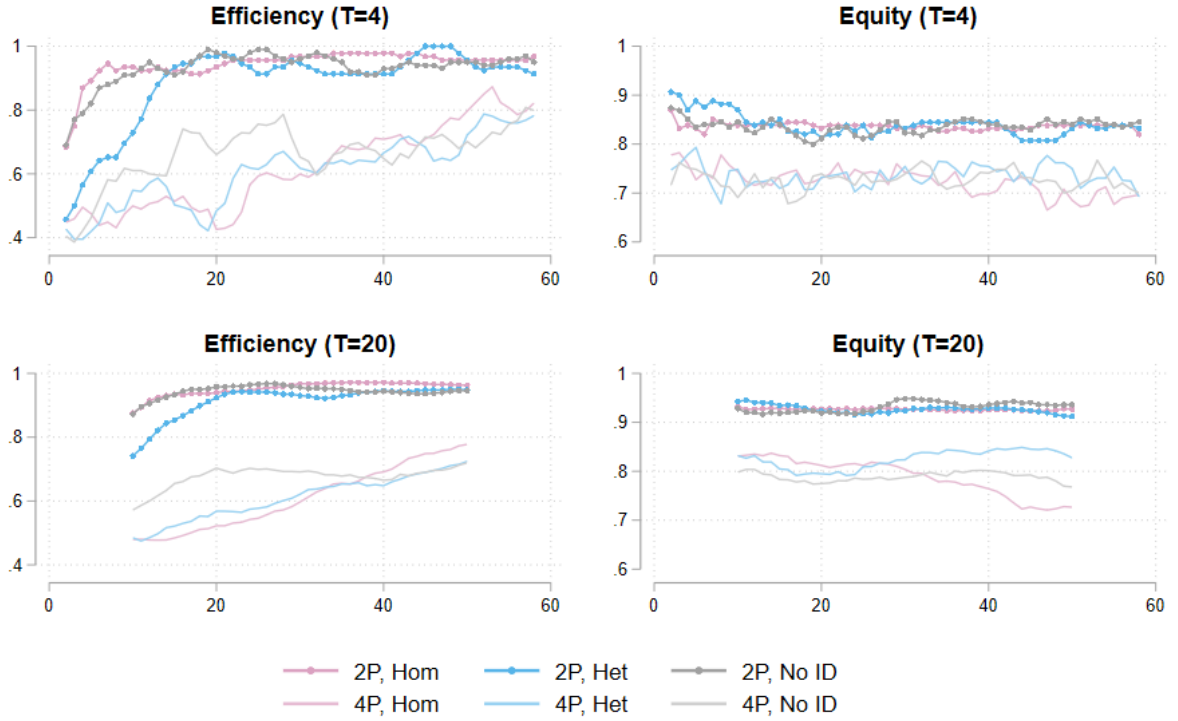
Efficiency and Equity are calculated here for the window $T = 20$ in the last block of 20 rounds. See also Figure A.1 for the window of all rounds, $T = 60$ and Figure A.2 for plots in each block of 20 rounds.

5.2 Effect of Group Size and Composition

Since stable coordination patterns take time to form following varied learning processes, we first examine and compare the evolution of efficiency and equity over time across the treatments. Figure 2 reveals that four-player groups learn to coordinate more slowly and perform consistently worse throughout the game, achieving lower efficiency and equity compared to two-player groups. This suggests that the strategic complexity introduced by larger group size plays a dominant role in hindering intertemporal coordination.

¹²That small and large groups show a marked variance in their frequency of switching in Table 4 suggests that the dynamics of turn-taking might vary across group structures. One possible measure which captures the dynamics of turn-taking is the estimated length of a switching cycle. If a participant switches N times, the average number of round before a switch is $\frac{60}{N+1}$. The length of a switching cycle can be estimated to be twice of this. The median length of a switching cycle is approximately 2 in small groups and 6 in large groups.

Figure 2: Efficiency and Equity over Treatments



Values are computed using a window of size $T = 4$ and $T = 20$. For $T = 4$, the value at time t is calculated as the average of 4 consecutive rounds, $(t - 1, t, t + 1, t + 2)$. For $T = 20$, the moving average at time t covers 20 rounds, $(t - 9, \dots, t, \dots, t + 10)$. Observations without a full window are omitted.

Among two-player groups, group composition affects early-stage efficiency. Heterogeneous groups have noticeably lower efficiency in the initial rounds, though this gap closes after the first 20 rounds. For large groups, identity composition has some effect on efficiency in initial rounds, but also on equity in the later stages. Four-player no-identity groups have somewhat higher efficiency early on, while four-player homogeneous groups show notably lower equity in the last 20 rounds. To further analyze these dynamics, we decompose the regressions of efficiency and equity into three 20-round blocks.

Effects of group size

Table 5 shows that four-player groups have a significant negative impact on efficiency and equity across all blocks ($p < 0.005$). In Block 1, larger group sizes lead to a 39.7 percentage point decline in efficiency and a 10.6 percentage point reduction in equity. Unlike two-player groups, which achieve an average efficiency above 0.9 shortly after Block 1, four-player groups take

Table 5: Efficiency and Equity Regressions

<i>Dep Var:</i>	Efficiency			Equity		
	(1) Block 1	(2) Block 2	(3) Block 3	(4) Block 1	(5) Block 2	(6) Block 3
No ID	-0.004 (0.056)	-0.011 (0.038)	-0.017 (0.046)	0.002 (0.039)	0.021 (0.039)	0.009 (0.043)
Het	-0.135** (0.060)	-0.037 (0.048)	-0.015 (0.045)	0.012 (0.035)	-0.004 (0.045)	-0.016 (0.048)
4P	-0.397*** (0.054)	-0.370*** (0.055)	-0.185*** (0.059)	-0.106*** (0.034)	-0.144*** (0.046)	-0.230*** (0.058)
No ID $\times$ 4P	0.096 (0.073)	0.106 (0.077)	-0.041 (0.085)	-0.044 (0.052)	-0.040 (0.060)	0.037 (0.075)
Het $\times$ 4P	0.140* (0.072)	0.061 (0.080)	-0.038 (0.081)	-0.023 (0.043)	0.024 (0.059)	0.127* (0.074)
Constant	0.876*** (0.042)	0.967*** (0.030)	0.963*** (0.035)	0.934*** (0.029)	0.947*** (0.034)	0.949*** (0.034)
<i>Implied differences in large groups</i>						
4P, No ID - 4P, Hom	0.092* (0.048)	0.094 (0.067)	-0.058 (0.071)	-0.043 (0.035)	-0.018 (0.046)	0.046 (0.062)
4P, Het - 4P, Hom	0.005 (0.041)	0.024 (0.063)	-0.053 (0.067)	-0.010 (0.025)	0.021 (0.038)	0.111* (0.057)
Observations	140	140	140	140	140	140

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-level OLS regressions with robust standard errors in parentheses. Group-level efficiency and equity are calculated over $T = 20$ rounds. See also Table A.1 in the appendix for results on averages over all rounds.

longer to learn how to coordinate and gradually improve their efficiency. By Block 3, the efficiency gap caused by group size shrinks to 18.5 percentage points. However, this improvement in efficiency appears to come at the cost of equity: four-player groups suffer a 23 percentage point drop in equity relative to two-player groups.

Effects of Group Composition

In contrast to the unambiguous effects of group size, effects of group composition are more nuanced and differ between two-player and four-player groups, and also for efficiency and equity.

With two players, heterogeneous groups exhibit noticeably lower efficiency, albeit temporarily in the first block. Specifically, in Block 1, heterogeneous groups experience a 13.5 percentage point drop in efficiency compared to homogeneous groups ($p = 0.026$). In contrast, no significant efficiency differences are observed between no-identity groups and homogeneous groups across all blocks. Similarly, equity does not significantly differ when comparing homogeneous with either heterogeneous or no identity groups.

With four players, identity composition has some temporary effects on efficiency but appears more relevant for equity later on. For efficiency, there is no significant difference between heterogeneous and homogeneous groups. No-identity groups show a modest 9.2 percentage point increase in efficiency in Block 1 relative to homogeneous groups ($p = 0.056$).¹³ For equity, differences emerge towards the end of the game when comparing heterogeneous and homogeneous groups. In Block 3, heterogeneous groups are 11.1 percentage points more equitable than homogeneous groups ($p = 0.053$).¹⁴ While no identity groups are also 4.6 percentage points more equitable than homogeneous groups in Block 3, this is not significant at conventional levels ($p = 0.459$).

5.3 Mechanisms

To understand why intertemporal coordination differs across groups, we first identify two behavioral mechanisms. Groups may differ in their (i) willingness to take turns and (ii) difficulty of sustaining coordination over time. Achieving both efficiency and equity requires members to alternate access to the high-payoff market, which involves short-run sacrifice in anticipation of future reciprocation. Even when members intend to take turns, attempts to do so may, however, be undermined by miscoordination on the timing of actions.

To capture these behavioral patterns, we construct two proxy measures. The first measures how frequently group members alternate between markets over a window of T rounds, reflecting attempts to take turns. The second measures instability in coordination patterns over the same window using the standard deviation of the proportion choosing Market A. Appendix C provides details of these measures.¹⁵ Table A.3 shows that these proxies explain a substantial portion of the variation in outcomes across the treatments. Turn-taking attempts are associated with both higher efficiency and higher equity, whereas coordination instability is associated primarily with lower efficiency.

These findings raise a further question: why do turn-taking patterns differ across groups and persist over time? We examine two potential explanations. The first, which we call *initial-*

¹³The difference in difference of the effects of heterogeneous identity composition in four-player groups relative to two-player groups is significant, $p = 0.055$ for Block 1. The difference in the difference of the effects of no identity composition in four-player groups relative to two-player groups is insignificant.

¹⁴The difference in the difference of the effects of heterogeneous identity composition in four-player groups relative to two-player groups is marginally significant, $p = 0.090$ for Block 3.

¹⁵While related, the two measures capture distinct aspects of behavior. A group may attempt to take turns but still fail to synchronize the timing of actions. Conversely, a group may coordinate on the timing of actions without engaging in turn-taking. Empirically, the correlation between the two measures is modest (Pearson $\rho = -0.388$), and both remain significantly affected by the treatments when controlling for the other measure (Table A.2).

outcome path dependence, arises when subsequent outcomes depend on whether groups happen to coordinate successfully in the initial round. The second, *behavioral path dependence*, arises when different behavioral patterns emerge within groups during early play, creating strategic frictions that shape subsequent outcomes. We examine each in turn.

Initial-outcome path dependence

If coincidental success of coordination in the initial round helps groups establish stable patterns of play, differences in subsequent performance may partly reflect dependence on this initial outcome. Since two-player groups exhibit higher rates of coordination in Round 1 (see Table 4), this could contribute to the observed group-size differences.

However, we find little evidence supporting this explanation. Supplementary analysis shows that coordination in Round 1 is not significantly associated with turn-taking attempts or coordination stability in later rounds (Table A.4, Columns 1 to 6). Likewise, Round 1 coordination does not significantly predict efficiency or equity in later rounds (Columns 7 to 12). Table A.5 also provides little evidence that the effects of Round 1 coordination on subsequent behavior and outcomes differ between two-player and four-player groups.¹⁶ These results suggest that persistent treatment differences are not driven by chance coordination in the initial round, while leaving open the possibility that behavioral patterns emerging later in Block 1 persist over time.

Behavioural path dependence

To examine how behavioral patterns develop within groups, we follow the approach of Bolle and Tan (2021) in classifying individuals by their proneness to conflict strategies, thus generating a typology of behavioral types. Using a k-means clustering procedure (Hartigan and Wong, 1979), we classify participants in each block according to their patterns of market entry within that block (see Appendix D for details). These classifications provide a more granular description of coordination dynamics than the aggregate measures of turn-taking attempts and coordination stability.¹⁷ Because they are derived from choices within the interaction, they may reflect fixed individual traits as well as responses to others and prior group dynamics.

¹⁶In a survivor analysis of which groups continue to fail at turn taking in Table A.6, we also find that Round 1 coordination does not have a significant impact. There are also no significant differences in the effects of Round 1 coordination by group size.

¹⁷Unlike settings where a small set of canonical dynamic strategies can be specified ex ante (e.g., Dal Bó and Fréchette, 2011), intertemporal coordination (or lack thereof) in our environment can arise from a wide range of dynamic strategies, including multiple patterns of turn-taking. This makes it difficult to precisely define such benchmark strategies a priori, motivating our use of a data-driven clustering approach.

This approach identifies four behavioral types with distinct behavioral strategies.¹⁸

1. **Perfect Turn-takers:** individuals who alternate almost perfectly between Market A and Market B.
2. **Imperfect Turn-takers:** individuals who alternate between markets less frequently than perfect turn-takers, consistent with slower or longer-cycle alternation.
3. **Exploiters:** individuals who predominantly choose the higher-payoff Market A rather than alternating.
4. **Compromisers:** individuals who predominantly choose the lower-payoff Market B rather than alternating.

The composition of behavioral types within groups, shown in Figure 3, helps explain differences in outcomes across and within group sizes. Groups with more perfect turn-takers exhibit stronger turn-taking attempts and more stable coordination patterns, the two behavioral proxies introduced earlier (see Table A.7). Consequently, these groups also achieve higher efficiency and equity, whereas other types are associated with poorer outcomes (see Table A.8).¹⁹

There are several notable patterns. In two-player groups, heterogeneous groups initially contain fewer perfect turn-takers (proportion test, $p = 0.101$) and more imperfect turn-takers (proportion test, $p = 0.002$) in the first block, corresponding to the slower emergence of coordination. In four-player groups, homogeneous groups exhibit a higher share of exploiters and compromisers relative to heterogeneous groups in the final block (proportion test, $p = 0.022$). These types coexist in 44 percent of cases where at least one player is an exploiter, helping explain why four-player homogeneous groups display lower equity despite similar efficiency: some members persistently enter the high-payoff market while others accommodate.

Finally, Figure 3 shows that these behavioral strategies are relatively persistent across blocks. Approximately 55% of participants retain the same classification throughout the game (see Table A.9). In particular, among those who adopt perfect turn-taking in one block, 92 to 94% retain the same strategy in the subsequent block. Early behavioral composition is also predictive of later outcomes. Groups with a higher proportion of turn-takers in Block 1 are more likely to achieve higher equity in Block 3 (Table A.10, Column 2), even after controlling for outcomes in Block

¹⁸The identified types mirror those found by Bolle and Tan (2021) in the Vendetta game (Bolle et al., 2014), a game of conflict where players can but should not steal from each other in equilibrium when they are in an equitable steady state, and stealing is inefficient. Our Turn-takers are akin to their “Gamesmen”, who play in equilibrium; Exploiters are like their “Sharks”, who persist in stealing; and Compromisers are akin to the “Meeks”, who give in to exploitation by settling for lower payoffs.

¹⁹Small groups have a significantly higher proportion of perfect turn-takers than large groups (proportion test, $p < 0.001$ in all blocks), consistent with their higher efficiency and equity.

	Block 1										Block 2										Block 3									
	2P					4P					Pooled					2P					4P					Pooled				
	Hom	Het	No ID	Hom	Het	No ID	Hom	Het	No ID	Hom	Het	No ID	Hom	Het	No ID	Hom	Het	No ID	Hom	Het	No ID	Hom	Het	No ID						
Perfect Turn-Taker	80.43	65.22	80.00	22.73	18.75	19.57	86.96	82.61	84.00	26.14	33.33	33.70	86.96	82.61	90.00	27.27	32.29	35.87	50.48											
Imperfect Turn-Taker	0.00	19.57	8.00	47.73	52.08	43.48	2.17	6.52	12.00	32.95	34.38	34.78	0.00	6.52	2.00	27.27	38.54	25.00	21.05											
Exploiter	15.22	15.22	10.00	26.14	19.79	28.26	6.52	8.70	4.00	29.55	23.96	23.91	8.70	8.70	6.00	27.27	20.83	23.91	18.42											
Compromiser	4.35	0.00	2.00	3.41	9.38	8.70	4.35	2.17	0.00	11.36	8.33	7.61	4.35	2.17	2.00	18.18	8.33	15.22	10.05											

Items in table indicate column percentages. Each row of squares in the figure for each treatment represent the members of each group. The individuals represented by the squares have fixed positions across the blocks such that it is possible to see the evolution of strategies of each individual over the blocks.

1. The predictive associations with subsequent efficiency and equity are also stronger among four-player groups (Columns 3 and 4).

Taken together, these findings highlight how the emergence and persistence of behavioral strategies are linked to efficiency and equity across groups. In smaller two-player groups, perfect turn-taking quickly becomes dominant, resulting in consistently efficient and equitable outcomes. Larger four-player groups instead exhibit greater behavioral heterogeneity, with multiple strategy types often coexisting. This diversity is associated with greater variation in subsequent outcomes: groups with more perfect turn-takers subsequently exhibit more turn-taking among other members and are more likely to achieve efficient and equitable outcomes, whereas groups with more exploiters and compromisers tend to converge toward efficient but unequal outcomes.

6 Discussion

In this study, we examine how group size and identity composition shape intertemporal coordination. Larger groups make coordination more difficult and are associated with a more diverse set of behavioral strategies, which in turn shape the dynamics and persistence of coordination. These findings suggest that improving coordination in complex environments may require shaping the strategies that emerge, for example through communication or the establishment of norms.

On group composition, effects depend on group size. In two-player groups, heterogeneous groups initially exhibit lower efficiency. In four-player groups, homogeneous groups exhibit greater inequity over time. These patterns suggest that the stronger cohesion associated with shared identity does not always promote intertemporal coordination. In larger, more complex environments, homogeneous groups may instead enable both exploitation and tolerance for it, whereas such inequities are less readily accommodated in heterogeneous groups. This is consistent with the idea that identity differences may provide coordination cues in more complex environments, making deviations more salient and less easily accommodated.

These findings have important implications for institutional design. Sustaining intertemporal cooperation depends not only on cohesion, but also on scale and coordination complexity. In larger or more complex environments, institutions must account for greater behavioral heterogeneity and more fragile coordination, rather than relying solely on shared identity. Understanding these constraints is central to designing systems that sustain efficiency and equity over time. Future research could examine how contextual features such as leadership or trust shape these dynamics, and whether tools such as pre-play communication ([Agranov et al., 2024](#)), commitment devices, or reputation systems can mitigate coordination frictions in larger groups.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT to improve grammar and clarity after writing the full draft. All substantive content, argumentation, experimental design, and framing reflect the authors' own thinking. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

- Agranov, M., Kim, J., and Yariv, L. (2024). Coordination with differential time preferences: Experimental evidence. *American Economic Review: Insights*, 6(4):543–557.
- Attanasi, G., Hopfensitz, A., Lorini, E., and Moisan, F. (2016). Social connectedness improves co-ordination on individually costly, efficient outcomes. *European Economic Review*, 90:86–106.
- Axelrod, R. (1984). *The Evolution of Cooperation*. Basic Books, New York.
- Barcelo, H. and Capraro, V. (2015). Group size effect on cooperation in one-shot social dilemmas. *Scientific Reports*, 5:7937.
- Bjedov, T., Madiès, T., and Villeval, M. C. (2016). Communication and Coordination in a Two-Stage Game. *Economic Inquiry*, 54(3):1519–1540.
- Bolle, F. and Tan, J. H. (2021). Behavioral types of the dark side: identifying heterogeneous conflict strategies. *Journal of the Economic Science Association*, 7(1):49–63.
- Bolle, F., Tan, J. H., and Zizzo, D. J. (2014). Vendettas. *American Economic Journal: Microeconomics*, 6(2):93–130.
- Bruttel, L. and Güth, W. (2018). Asymmetric voluntary cooperation: a repeated sequential best-shot experiment. *International Journal of Game Theory*, 47(3):873–891.
- Camerer, C. F. and Lovallo, D. (1999). Overconfidence and excess entry: An experimental approach. *American Economic Review*, 89(1):306–318.
- Cason, T. N., Lau, S.-H. P., and Mui, V.-L. (2013). Learning, teaching, and turn taking in the repeated assignment game. *Economic Theory*, 54(2):335–357.

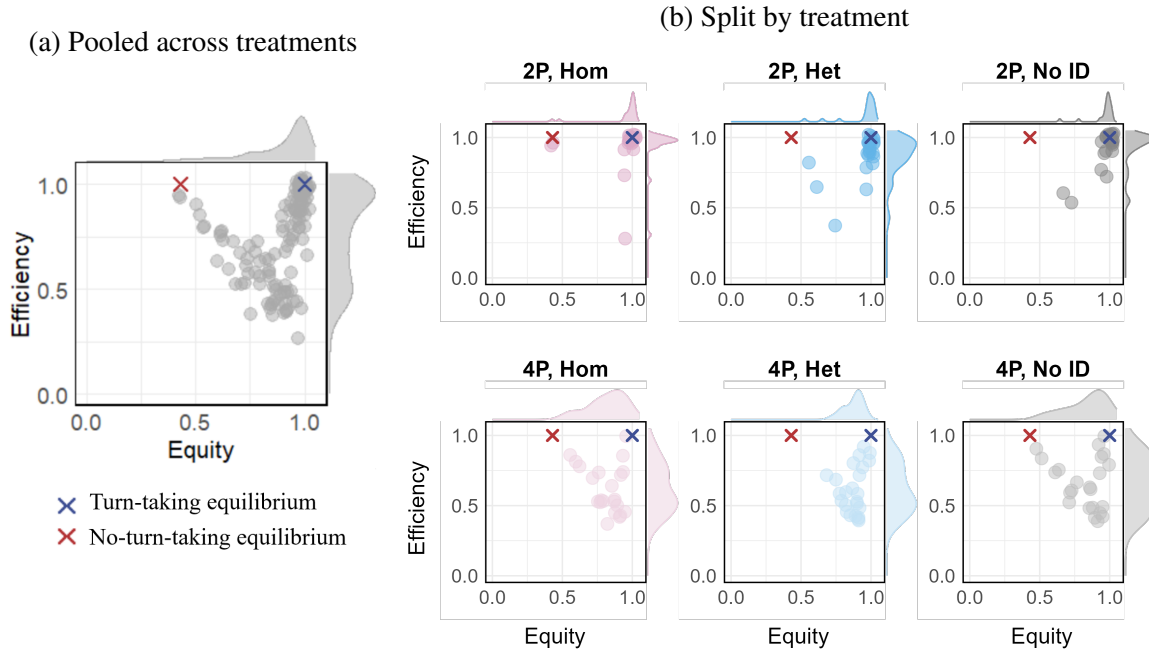
- Charness, G. and Rabin, M. (2002). Understanding social preferences with simple tests*. *The Quarterly Journal of Economics*, 117(3):817–869.
- Charness, G., Rigotti, L., and Rustichini, A. (2007). Individual behavior and group membership. *American Economic Review*, 97(4):1340–1352.
- Chen, R. and Chen, Y. (2011). The potential of social identity for equilibrium selection. *American Economic Review*, 101(6):2562–2589.
- Chen, R., Chen, Y., and Riyanto, Y. E. (2021). Best practices in replication: a case study of common information in coordination games. *Experimental Economics*, 24(1):2–30.
- Chen, Y. and Li, S. (2009). Group identity and social preferences. *American Economic Review*, 99:431–457.
- Chen, Y., Li, S. X., Liu, T. X., and Shih, M. (2014). Which hat to wear? Impact of natural identities on coordination and cooperation. *Games and Economic Behavior*, 84:58–86.
- Chuah, S. H., Gächter, S., Hoffmann, R., and Tan, J. H. (2016). Religion, discrimination and trust across three cultures. *European Economic Review*, 90:280–301.
- Dal Bó, P. and Fréchet, G. R. (2011). The evolution of cooperation in infinitely repeated games: Experimental evidence. *American Economic Review*, 101(1):411–429.
- Dunbar, R. I. M. (1992). Neocortex size as a constraint on group size in primates. *Journal of Human Evolution*, 22(6):469–493.
- Eckel, C. C. and Grossman, P. J. (2005). Managing diversity by creating team identity. *Journal of Economic Behavior & Organization*, 58(3):371–392.
- Fehr, E. and Gächter, S. (2000). Cooperation and punishment in public goods experiments. *American Economic Review*, 90(4):980–994.
- Fehr, E. and Schmidt, K. M. (1999). A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics*, 114(3):817–868.
- Fischbacher, U. (2007). Z-Tree: Zurich toolbox for ready-made economic experiments. *Experimental Economics*, 10(2):171–178.

- Goette, L., Huffman, D., and Meier, S. (2006). The impact of group membership on cooperation and norm enforcement: Evidence using random assignment to real social groups. *American Economic Review*, 96(2):212–216.
- Hardin, G. (1968). The tragedy of the commons. *Science*, 162(3859):1243–1248.
- Hartigan, J. A. and Wong, M. A. (1979). A k-means clustering algorithm. *Applied Statistics*, 28(1):100–108.
- Huck, S., Normann, H.-T., and Oechssler, J. (2004). Two are few and four are many: number effects in experimental oligopolies. *Journal of Economic Behavior & Organization*, 53(4):435–446.
- Isaac, R. M., Walker, J. M., and Williams, A. W. (1994). Group size and the voluntary provision of public goods: Experimental evidence utilizing large groups. *Journal of Public Economics*, 54(1):1–36.
- Kaplan, T. R. and Ruffle, B. J. (2012). Which way to cooperate. *The Economic Journal*, 122(563):1042–1068.
- Knez, M. and Camerer, C. (1994). Creating expectational assets in the laboratory: Coordination in ‘weakest-link’ games. *Strategic Management Journal*, 15(S1):101–119.
- Kuzmics, C., Palfrey, T., and Rogers, B. W. (2014). Symmetric play in repeated allocation games. *Journal of Economic Theory*, 154:25–67.
- Lane, T. (2016). Discrimination in the laboratory: A meta-analysis of economics experiments. *European Economic Review*, 90:375–402.
- Lau, S.-H. P. and Mui, V.-L. (2008). Using turn taking to mitigate coordination and conflict problems in the repeated battle of the sexes game. *Theory and Decision*, 65:153–183.
- Lau, S.-H. P. and Mui, V.-L. (2012). Using turn taking to achieve intertemporal cooperation and symmetry in infinitely repeated 2×2 games. *Theory and Decision*, 72:167–188.
- Ledyard, J. O. (1995). Public goods: A survey of experimental research. In Kagel, J. H. and Roth, A. E., editors, *The Handbook of Experimental Economics*, pages 111–194. Princeton University Press, Princeton, NJ.

- Li, S. X. (2020). Group identity, ingroup favoritism, and discrimination. *Handbook of Labor, Human Resources and Population Economics*, pages 1–28.
- Nosenzo, D., Quercia, S., and Sefton, M. (2015). Cooperation in small groups: the effect of group size. *Experimental Economics*, 18(1):4–14.
- Olson, M. (1971). *The Logic of Collective Action: Public Goods and the Theory of Groups*, volume 124 of *Harvard Economic Studies*. Harvard University Press, Cambridge, Mass., revised edition.
- Ostrom, E. (1990). *Governing the commons: The evolution of institutions for collective action*. Cambridge University Press.
- Rabin, M. (1993). Incorporating fairness into game theory and economics. *American Economic Review*, 83(5):1281–1302.
- Riyanto, Y. E. and Roy, N. (2019). Path of intertemporal cooperation and limits to turn-taking behavior. *Journal of Economic Behavior & Organization*, 165:21–36.
- Smaldino, P. E. (2019). The evolution of the social self: Multidimensionality of social identity solves the coordination problems of a society. In *Beyond the Meme, Development and Structure in Cultural Evolution*, pages 445–469. University of Minnesota Press.
- Tajfel, H., Billig, M. G., Bundy, R. P., and Flament, C. (1971). Social categorization and intergroup behaviour. *European Journal of Social Psychology*, 1(2):149–178.
- Tajfel, H. and Turner, J. C. (1979). An integrative theory of intergroup conflict. *The Social Psychology of Intergroup Relations*, 33(47):74.
- Van Huyck, J. B., Battalio, R. C., and Beil, R. O. (1990). Tacit coordination games, strategic uncertainty, and coordination failure. *American Economic Review*, 80(1):234–248.
- Weber, R. A. (2006). Managing growth to achieve efficient coordination in large groups. *American Economic Review*, 96(1):114–126.
- Weimann, J., Brosig-Koch, J., Heinrich, T., Hennig-Schmidt, H., and Keser, C. (2019). Public good provision by large groups – the logic of collective action revisited. *European Economic Review*, 118:348–363.
- Zillante, A. (2017). Recognizing and Playing Turn Taking Strategies.

A Supplementary Results

Figure A.1: Efficiency and Equity (All Rounds, $T=60$)



[†] Efficiency and Equity are calculated here for the window $T = 60$.

Figure A.2: Evolution of Equity and Efficiency

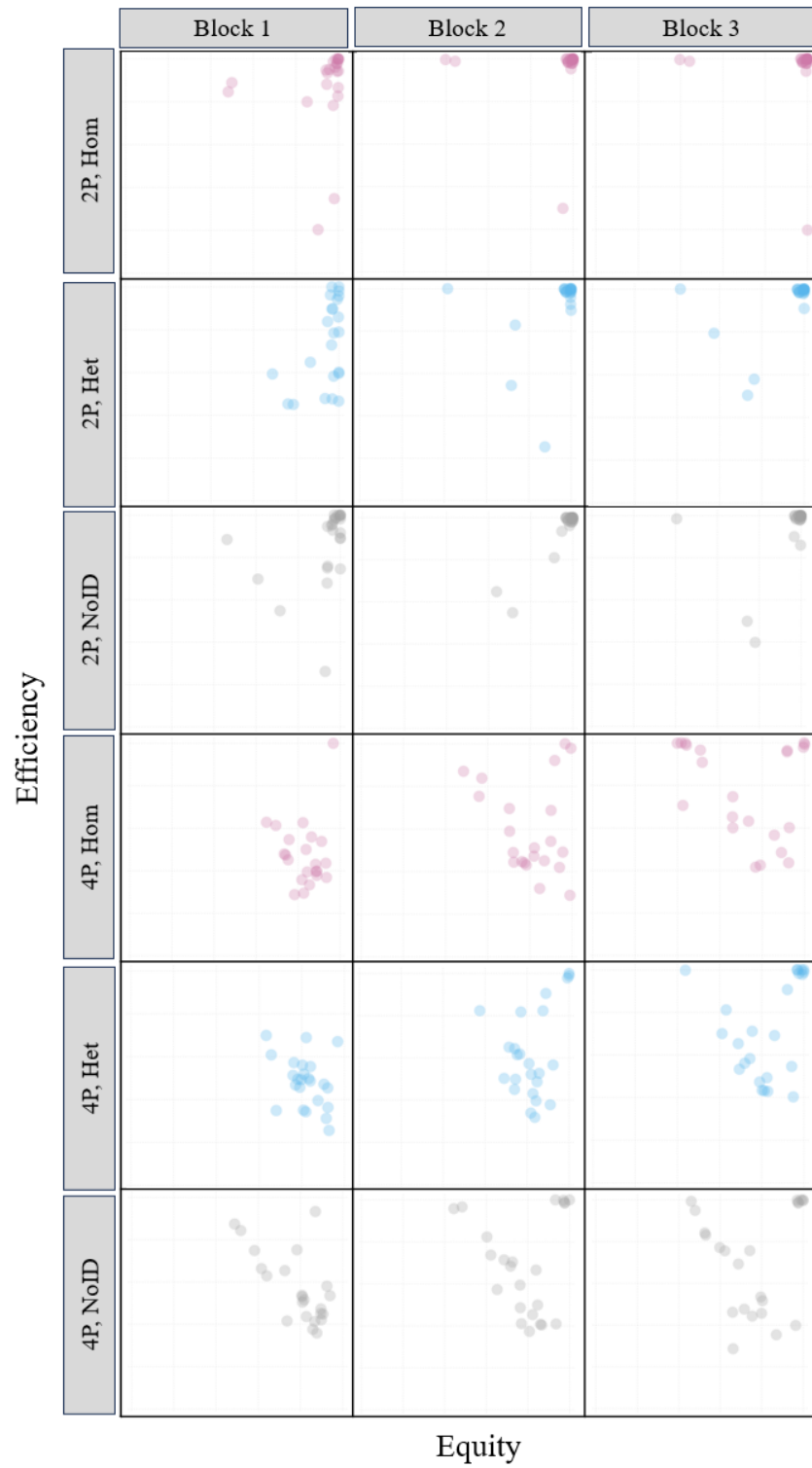


Table A.1: Regressions of Outcomes ($T = 20$) across all blocks

	(1) Efficiency	(2) Equity
No ID	-0.011 (0.041)	0.011 (0.038)
Het	-0.062 (0.043)	-0.002 (0.041)
4P	-0.318*** (0.047)	-0.160*** (0.041)
No ID $\times$ 4P	0.054 (0.065)	-0.016 (0.056)
Het $\times$ 4P	0.054 (0.065)	0.043 (0.052)
Constant	0.936*** (0.032)	0.943*** (0.032)
<i>Implied differences in large groups</i>		
4P, No ID - 4P, Hom	0.043 (0.051)	-0.005 (0.040)
4P, Het - 4P, Hom	-0.008 (0.048)	0.040 (0.031)
Observations	420	420

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-block level OLS regressions with group clustered standard errors in parentheses. Group-level Efficiency and Equity are calculated for each consecutive block of $T = 20$ rounds. With 60 rounds in total, there are 3 observations for each group in one column.

Table A.2: Turn-taking Attempts and Coordination Instability

<i>Dep Var:</i>	Group Attempts to Take Turns			Coordination Instability		
	(1) Block1	(2) Block2	(3) Block3	(4) Block1	(5) Block2	(6) Block3
No ID	0.025 (0.069)	0.083 (0.063)	0.035 (0.074)	0.009 (0.025)	0.028 (0.021)	0.015 (0.024)
Het	0.091 (0.070)	0.020 (0.079)	0.002 (0.078)	0.099*** (0.027)	0.028 (0.020)	0.014 (0.024)
4P	-0.073 (0.076)	-0.215** (0.097)	-0.327*** (0.102)	0.081*** (0.022)	0.105*** (0.028)	0.035 (0.030)
No ID $\times$ 4P	-0.113 (0.094)	0.009 (0.109)	0.097 (0.127)	-0.037 (0.031)	-0.043 (0.033)	0.015 (0.036)
Het $\times$ 4P	-0.066 (0.091)	0.088 (0.117)	0.163 (0.124)	-0.078** (0.034)	-0.020 (0.031)	0.009 (0.035)
Coordination Instability	-0.841*** (0.198)	-1.036*** (0.264)	-1.102*** (0.265)			
Group Attempts to Take Turns				-0.114*** (0.029)	-0.093*** (0.035)	-0.091*** (0.030)
Constant	0.922*** (0.047)	0.903*** (0.062)	0.919*** (0.062)	0.206*** (0.028)	0.100** (0.039)	0.104*** (0.038)
<i>Implied differences in large groups</i>						
4P, No ID - 4P, Hom	-0.088 (0.063)	0.092 (0.089)	0.132 (0.102)	-0.028 (0.019)	-0.015 (0.026)	0.030 (0.026)
4P, Het - 4P, Hom	0.025 (0.063)	0.107 (0.086)	0.165* (0.096)	0.021 (0.020)	0.008 (0.023)	0.024 (0.025)
Observations	140	140	140	140	140	140

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns 1 to 6: Group-level OLS regressions with robust standard errors in parentheses. Group Attempts to Take Turns is the average of individual attempts to take turns within a group in each window. Group-level turn-taking attempts and group-level coordination instability are calculated over $T = 20$ rounds in each column.

Table A.3: Mechanisms

Dep Var:	Efficiency						Equity					
	(1) Block1	(2) Block2	(3) Block3	(4) Block1	(5) Block2	(6) Block3	(7) Block1	(8) Block2	(9) Block3	(10) Block1	(11) Block2	(12) Block3
No ID	-0.004 (0.056)	-0.011 (0.038)	-0.017 (0.046)	0.001 (0.034)	0.028 (0.018)	0.009 (0.015)	0.002 (0.039)	0.021 (0.039)	0.009 (0.043)	-0.006 (0.021)	-0.002 (0.014)	-0.004 (0.009)
Het	-0.135** (0.060)	-0.037 (0.048)	-0.015 (0.045)	0.004 (0.034)	0.021 (0.022)	0.017 (0.017)	0.012 (0.035)	-0.004 (0.045)	-0.016 (0.048)	0.004 (0.022)	0.001 (0.018)	-0.012 (0.010)
4P	-0.397*** (0.054)	-0.370*** (0.055)	-0.185*** (0.059)	-0.224*** (0.042)	-0.078** (0.033)	-0.025 (0.023)	-0.106*** (0.034)	-0.144*** (0.046)	-0.230*** (0.058)	-0.056* (0.031)	0.002 (0.037)	-0.039 (0.031)
No ID × 4P	0.096 (0.073)	0.106 (0.077)	-0.041 (0.085)	0.076 (0.046)	0.009 (0.035)	-0.032 (0.038)	-0.044 (0.052)	-0.040 (0.060)	0.037 (0.075)	-0.011 (0.037)	-0.064* (0.036)	-0.010 (0.040)
Het × 4P	0.140* (0.072)	0.061 (0.080)	-0.038 (0.081)	0.028 (0.043)	-0.007 (0.038)	-0.058* (0.034)	-0.023 (0.043)	0.024 (0.059)	0.127* (0.074)	-0.018 (0.034)	-0.024 (0.039)	0.043 (0.036)
Group Attempts to Take Turns				0.191*** (0.068)	0.066* (0.038)	0.043 (0.034)				0.353*** (0.053)	0.403*** (0.040)	0.506*** (0.023)
Coordination Instability				-1.437*** (0.107)	-1.938*** (0.133)	-1.982*** (0.099)				0.059 (0.080)	-0.005 (0.108)	0.205* (0.104)
Constant	0.876*** (0.042)	0.967*** (0.030)	0.963*** (0.035)	0.878*** (0.073)	0.944*** (0.036)	0.970*** (0.032)	0.934*** (0.029)	0.947*** (0.034)	0.949*** (0.034)	0.635*** (0.058)	0.590*** (0.044)	0.492*** (0.023)
Implied differences in large groups												
4P, No ID - 4P, Hom	0.092* (0.048)	0.094 (0.067)	-0.058 (0.071)	0.077** (0.031)	0.037 (0.030)	-0.023 (0.035)	-0.043 (0.035)	-0.018 (0.046)	0.046 (0.062)	-0.016 (0.031)	-0.066** (0.033)	-0.014 (0.039)
4P, Het - 4P, Hom	0.005 (0.041)	0.024 (0.063)	-0.053 (0.067)	0.032 (0.027)	0.014 (0.031)	-0.041 (0.030)	-0.010 (0.025)	0.021 (0.038)	0.111* (0.057)	-0.014 (0.026)	-0.023 (0.035)	0.031 (0.035)
Observations	140	140	140	140	140	140	140	140	140	140	140	140

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-level OLS regressions with robust standard errors in parentheses. Group Attempts to Take Turns is the average of individual attempts to take turns within a group in each window. All variables are calculated over $T = 20$ rounds in each column.

Table A.4: The Influence of Early Path Dependence on Mechanisms and Outcomes

Dep Var:	Group Attempts to Take Turns			Coordination Instability			Efficiency		Equity			
	(1) Block 1	(2) Block 2	(3) Block 3	(4) Block 1	(5) Block 2	(6) Block 3	(7) Block 1	(8) Block 2	(9) Block 3	(10) Block 1	(11) Block 2	(12) Block 3
No ID	0.018 (0.087)	0.063 (0.072)	0.024 (0.077)	0.013 (0.029)	0.022 (0.023)	0.013 (0.025)	-0.009 (0.058)	-0.011 (0.038)	-0.012 (0.047)	0.002 (0.043)	0.024 (0.039)	0.010 (0.043)
Het	0.019 (0.079)	-0.005 (0.089)	-0.010 (0.088)	0.118*** (0.033)	0.029 (0.022)	0.015 (0.025)	-0.136** (0.062)	-0.037 (0.047)	-0.009 (0.044)	0.011 (0.040)	-0.000 (0.047)	-0.014 (0.050)
4P	-0.165** (0.083)	-0.346*** (0.095)	-0.395*** (0.104)	0.140*** (0.024)	0.138*** (0.020)	0.070*** (0.025)	-0.421*** (0.055)	-0.370*** (0.055)	-0.171*** (0.058)	-0.116*** (0.041)	-0.136*** (0.049)	-0.226*** (0.061)
No ID × 4P	-0.080 (0.107)	0.060 (0.113)	0.090 (0.129)	-0.042 (0.034)	-0.048 (0.034)	0.006 (0.036)	0.117 (0.078)	0.106 (0.077)	-0.040 (0.085)	-0.021 (0.055)	-0.040 (0.060)	0.037 (0.075)
Het × 4P	-0.008 (0.102)	0.108 (0.127)	0.160 (0.136)	-0.092** (0.038)	-0.031 (0.032)	-0.004 (0.035)	0.130* (0.074)	0.061 (0.079)	-0.052 (0.079)	-0.012 (0.049)	0.018 (0.062)	0.123 (0.077)
Coordinated in Round 1	0.052 (0.045)	0.043 (0.050)	0.036 (0.057)	-0.018 (0.016)	-0.003 (0.015)	-0.008 (0.015)	0.015 (0.033)	0.002 (0.033)	0.051 (0.033)	0.031 (0.023)	0.025 (0.025)	0.013 (0.031)
Constant	0.807*** (0.078)	0.859*** (0.079)	0.871*** (0.080)	0.072*** (0.021)	0.020 (0.015)	0.028 (0.018)	0.901*** (0.043)	0.966*** (0.032)	0.932*** (0.037)	0.922*** (0.041)	0.931*** (0.044)	0.941*** (0.046)
Implied differences in large groups												
4P, No ID - 4P, Hom	-0.062 (0.063)	0.123 (0.088)	0.114 (0.104)	-0.029 (0.018)	-0.026 (0.025)	0.019 (0.026)	0.107** (0.051)	0.095 (0.067)	-0.053 (0.070)	-0.019 (0.034)	-0.016 (0.046)	0.047 (0.062)
4P, Het - 4P, Hom	0.011 (0.062)	0.103 (0.088)	0.149 (0.100)	0.025 (0.018)	-0.001 (0.022)	0.011 (0.025)	-0.006 (0.041)	0.024 (0.064)	-0.061 (0.066)	-0.001 (0.026)	0.017 (0.038)	0.109* (0.057)
Observations	140	140	140	140	140	140	140	140	140	140	140	140

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-level OLS regressions with robust standard errors in parentheses. For Blocks 2 and 3, all dependent variables are calculated at the group level over $T = 20$ rounds. For Block 1, given that coordination in Round 1 is used as an independent variable, we exclude Rounds 1 and 2 from the calculation of dependent variables to obtain an even window of $T = 18$.

Table A.5: The Influence of Early Path Dependence on Mechanisms and Outcomes (with interaction)

Dep Var:	Group Attempts to Take Turns			Coordination Instability			Efficiency			Equity		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Block 1	Block 2	Block 3	Block 1	Block 2	Block 3	Block 1	Block 2	Block 3	Block 1	Block 2	Block 3
No ID	0.021 (0.086)	0.064 (0.073)	0.026 (0.078)	0.013 (0.030)	0.023 (0.023)	0.014 (0.025)	-0.010 (0.057)	-0.012 (0.038)	-0.017 (0.045)	0.004 (0.043)	0.024 (0.040)	0.012 (0.043)
Het	0.024 (0.081)	-0.004 (0.091)	-0.007 (0.091)	0.118*** (0.033)	0.030 (0.022)	0.017 (0.025)	-0.137*** (0.061)	-0.038 (0.046)	-0.015 (0.042)	0.013 (0.041)	-0.000 (0.049)	-0.011 (0.052)
4P	-0.124 (0.097)	-0.339*** (0.112)	-0.366*** (0.120)	0.140*** (0.027)	0.145*** (0.022)	0.088*** (0.025)	-0.433*** (0.051)	-0.380*** (0.056)	-0.222*** (0.061)	-0.099* (0.052)	-0.136*** (0.061)	-0.200*** (0.072)
No ID × 4P	-0.088 (0.106)	0.058 (0.114)	0.084 (0.130)	-0.042 (0.035)	-0.050 (0.034)	0.002 (0.036)	0.119 (0.076)	0.108 (0.078)	-0.029 (0.084)	-0.025 (0.055)	-0.040 (0.061)	0.032 (0.076)
Het × 4P	-0.006 (0.102)	0.109 (0.127)	0.161 (0.135)	-0.092*** (0.038)	-0.030 (0.032)	-0.003 (0.035)	0.129* (0.074)	0.060 (0.080)	-0.054 (0.078)	-0.011 (0.049)	0.018 (0.062)	0.124 (0.076)
Coordinated in Round 1	0.093 (0.067)	0.050 (0.066)	0.066 (0.071)	-0.018 (0.027)	0.004 (0.019)	0.011 (0.020)	0.003 (0.046)	-0.009 (0.033)	-0.000 (0.034)	0.049 (0.036)	0.026 (0.036)	0.039 (0.040)
Coordinated in Round 1 × 4P	-0.091 (0.088)	-0.016 (0.100)	-0.065 (0.116)	0.001 (0.032)	-0.015 (0.030)	-0.041 (0.031)	0.027 (0.065)	0.023 (0.069)	0.113* (0.068)	-0.040 (0.044)	-0.001 (0.049)	-0.057 (0.064)
Constant	0.781*** (0.088)	0.854*** (0.091)	0.853*** (0.091)	0.072*** (0.024)	0.015 (0.014)	0.016 (0.017)	0.909*** (0.041)	0.973*** (0.028)	0.963*** (0.030)	0.911*** (0.049)	0.931*** (0.051)	0.925*** (0.052)
Implied differences in large groups												
4P, No ID - 4P, Hom	-0.067 (0.063)	0.122 (0.088)	0.110 (0.104)	-0.029 (0.018)	-0.027 (0.026)	0.017 (0.026)	0.109*** (0.051)	0.096 (0.068)	-0.046 (0.071)	-0.021 (0.035)	-0.016 (0.046)	0.044 (0.062)
4P, Het - 4P, Hom	0.018 (0.062)	0.104 (0.088)	0.154 (0.100)	0.025 (0.018)	-0.000 (0.023)	0.014 (0.024)	-0.008 (0.042)	0.022 (0.065)	-0.069 (0.066)	0.002 (0.026)	0.017 (0.038)	0.113*** (0.056)
Observations	140	140	140	140	140	140	140	140	140	140	140	140

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-level OLS regressions with robust standard errors in parentheses. For Blocks 2 and 3, all dependent variables are calculated at the group level over $T = 20$ rounds. For Block 1, given that coordination in Round 1 is used as an independent variable, we exclude Rounds 1 and 2 from the calculation of dependent variables to obtain an even window of $T = 18$.

Table A.6: Estimates from Survival Analysis

<i>Log hazard ratio of achieving turn taking</i>	(1) Exponential	(2) Weibull
Coordinated in Round 1	0.244 (0.507)	0.197 (0.347)
4P	-2.786*** (0.464)	-2.441*** (0.396)
Coordinated in Round 1 $\times$ 4P	0.231 (0.700)	0.261 (0.575)
Constant	-2.182*** (0.347)	-1.128*** (0.219)
Observations	2789	2789

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Proportional hazards model with group clustered standard errors in parentheses. The analysis uses group-round-level data up to the first round in which successful turn taking occurs. Successful turn taking is defined as both future efficiency and future equity over the subsequent 20 rounds being at least 0.8.

Table A.7: Types, Attempts to Take Turns and Coordination Instability

<i>Dep Var:</i>	Group Attempts to Take Turns				Coordination Instability			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Prop. Exploiter	-1.106*** (0.083)				0.214*** (0.023)			
Prop. Compromiser		-1.728*** (0.064)				0.008 (0.037)		
Prop. Perfect TT			0.541*** (0.026)				-0.151*** (0.010)	
Prop. Imperfect TT				-0.305*** (0.057)				0.233*** (0.016)
Constant	0.933*** (0.010)	0.853*** (0.016)	0.447*** (0.025)	0.813*** (0.027)	0.076*** (0.007)	0.112*** (0.008)	0.196*** (0.009)	0.062*** (0.005)
Observations	420	420	420	420	420	420	420	420

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-block-level OLS regressions with group clustered standard errors in parentheses. Group Attempts to Take Turns is the average of individual attempts to take turns within a group in each window. Group-level turn-taking attempts and group-level coordination instability are calculated for each consecutive block of $T = 20$ rounds. With 60 rounds in total, there are 3 observations for each group in one column.

Table A.8: Behavioral Types and Group Efficiency/Equity

<i>Dep Var:</i>	Efficiency				Equity			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Prop. Exploiter	-0.622*** (0.052)				-0.439*** (0.062)			
Prop. Compromiser		-0.102 (0.103)				-0.896*** (0.040)		
Prop. Perfect TT			0.400*** (0.023)				0.245*** (0.016)	
Prop. Imperfect TT				-0.568*** (0.047)				-0.152*** (0.030)
Constant	0.877*** (0.015)	0.779*** (0.020)	0.551*** (0.022)	0.896*** (0.012)	0.946*** (0.008)	0.927*** (0.008)	0.736*** (0.016)	0.905*** (0.015)
Observations	420	420	420	420	420	420	420	420

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-block-level OLS regressions with group-clustered standard errors in parentheses. Group-level efficiency and equity are calculated for each consecutive block of $T = 20$ rounds. With 60 rounds in total, there are 3 observations for each group in one column.

Table A.9: Transition Matrix for Individual Strategy Types

Type in Block 2:					
	Perfect Turn-Taker	Imperfect Turn-Taker	Exploiter	Compromiser	Total
Type in Block 1:					
Perfect Turn-Taker	150 (92.02%)	7 (4.29%)	4 (2.45%)	2 (1.23%)	163 (100.00%)
Imperfect Turn-Taker	30 (20.69%)	72 (49.66%)	30 (20.69%)	13 (8.97%)	145 (100.00%)
Exploiter	23 (26.44%)	19 (21.84%)	45 (51.72%)	0 (0.00%)	87 (100.00%)
Compromiser	3 (13.04%)	6 (26.09%)	1 (4.35%)	13 (56.52%)	23 (100.00%)
Total	206	104	80	28	418
Type in Block 3:					
	Perfect Turn-Taker	Imperfect Turn-Taker	Exploiter	Compromiser	Total
Type in Block 2:					
Perfect Turn-Taker	194 (94.17%)	7 (3.40%)	2 (0.97%)	3 (1.46%)	206 (100.00%)
Imperfect Turn-Taker	13 (12.50%)	58 (55.77%)	15 (14.42%)	18 (17.31%)	104 (100.00%)
Exploiter	3 (3.75%)	16 (20.00%)	58 (72.50%)	3 (3.75%)	80 (100.00%)
Compromiser	1 (3.57%)	7 (25.00%)	2 (7.14%)	18 (64.29%)	28 (100.00%)
Total	211	88	77	42	418

Items in the table show the frequency of individuals in each type across two consecutive blocks, with row percentages in parentheses. For example, out of 163 Perfect Turn-Takers in Block 1, 150 of them are still classified as Perfect Turn-Taker in Block 2, while 7 of them transition to Imperfect Turn-Taker, 4 to Exploiter, and 2 to Compromiser.

Table A.10: Early Type Composition and Block 3 Performance

<i>Dep Var:</i>	Block 3:			
	(1) Efficiency	(2) Equity	(3) Efficiency	(4) Equity
No ID	-0.015 (0.040)	0.007 (0.017)	-0.016 (0.041)	0.007 (0.012)
Het	0.045 (0.050)	-0.022 (0.024)	0.037 (0.051)	-0.034* (0.020)
4P	0.014 (0.085)	-0.121*** (0.046)	-0.047 (0.107)	-0.219*** (0.048)
No ID $\times$ 4P	-0.071 (0.080)	0.099** (0.048)	-0.067 (0.082)	0.105** (0.043)
Het $\times$ 4P	-0.097 (0.079)	0.151*** (0.053)	-0.084 (0.082)	0.171*** (0.049)
Prop. Perfect TT in Block 1	0.060 (0.077)	0.249*** (0.060)	-0.026 (0.089)	0.113** (0.053)
4P $\times$ Prop. Perfect TT in Block 1			0.191* (0.114)	0.305*** (0.068)
Group Equity in Block 1	0.113 (0.203)	0.659*** (0.120)	0.224 (0.215)	0.837*** (0.110)
Group Efficiency in Block 1	0.386** (0.164)	-0.264*** (0.090)	0.435*** (0.157)	-0.186** (0.080)
Constant	0.471** (0.234)	0.365*** (0.135)	0.392* (0.228)	0.239* (0.123)
<i>Implied coefficient for large groups</i>				
Prop. Perfect TT in Block 1			0.166* (0.100)	0.418*** (0.065)
Observations	140	140	140	140

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-level OLS regressions with robust standard errors in parentheses. Group-level Efficiency and Equity are calculated over $T = 20$ rounds in each column.

B Robustness Checks for Shorter Term Window

In Table B.1, we present results on efficiency and equity, which are measured over shorter windows of $T = 4$. The effects of group size are similar to the findings in Table 5. Efficiency (Block 1 and Block 2: $p < 0.001$; Block 3: $p = 0.002$) and equity (all blocks: $p < 0.001$) are significantly higher in small groups compared to large groups. In small groups, heterogeneous identity composition leads to lower efficiency compared to homogeneous groups in Block 1 ($p = 0.024$), consistent with the findings using longer windows of $T = 20$ in Table 5. However, in large groups, there is no significant variation in equity by group composition when using short windows ($T = 4$), in contrast to the low equity observed in large homogeneous groups under longer windows ($T = 20$). This suggests that the differences in equity are primarily driven by longer-term dynamics.

Table B.2 reveals a similar pattern to Table A.2 and aligns with the efficiency and equity results presented in Table B.1. Large groups have lower attempts to take turns (Block 1: $p = 0.006$; Block 2 and Block 3: $p < 0.001$). Coordination Instability is higher in large groups in Block 1 and Block 2 compared to small groups ($p < 0.001$), while no significant difference is observed in Block 3, suggesting that large groups eventually develop a relatively stable, albeit inefficient, coordination pattern.

In small groups, coordination instability is noticeably higher in Block 1 when members have heterogeneous identities ($p = 0.001$), which corresponds to the lower efficiency observed in such groups. In large groups, turn-taking attempts are still lower in homogeneous groups in Block 3, but the differences with no-identity groups and heterogeneous groups become weaker when using shorter-term windows ($p = 0.264$ and $p = 0.397$, respectively) compared to Table A.2 which uses longer-term windows. This further supports the idea that variations in equity are primarily driven by longer-term dynamics.

Table B.1: Regressions of Efficiency and Equity (T=4)

	Efficiency			Equity		
	(1) Block1	(2) Block2	(3) Block3	(4) Block1	(5) Block2	(6) Block3
No ID	-0.004 (0.055)	-0.011 (0.038)	-0.017 (0.046)	0.001 (0.021)	-0.001 (0.022)	0.006 (0.021)
Het	-0.135** (0.059)	-0.037 (0.047)	-0.015 (0.044)	0.020 (0.018)	-0.001 (0.023)	-0.006 (0.023)
4P	-0.397*** (0.053)	-0.370*** (0.054)	-0.185*** (0.058)	-0.103*** (0.019)	-0.101*** (0.024)	-0.128*** (0.028)
No ID $\times$ 4P	0.096 (0.072)	0.106 (0.076)	-0.041 (0.083)	-0.027 (0.027)	-0.001 (0.033)	0.011 (0.036)
Het $\times$ 4P	0.140* (0.071)	0.061 (0.078)	-0.038 (0.080)	-0.024 (0.024)	0.016 (0.032)	0.036 (0.037)
Constant	0.876*** (0.042)	0.967*** (0.030)	0.963*** (0.034)	0.842*** (0.016)	0.832*** (0.017)	0.834*** (0.017)
Observations	700	700	700	700	700	700
<i>Implied differences in large groups</i>						
4P, No ID – 4P, Hom	0.092* (0.047)	0.094 (0.066)	-0.058 (0.070)	-0.025 (0.017)	-0.002 (0.025)	0.017 (0.030)
4P, Het – 4P, Hom	0.005 (0.040)	0.024 (0.063)	-0.053 (0.066)	-0.004 (0.015)	0.015 (0.023)	0.030 (0.029)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Group-Window-level OLS regressions with group clustered standard errors in parentheses. Group-level efficiency/equity are calculated for every 4 consecutive rounds. Since each block consists of 20 rounds, there are 5 observations for every group in each column.

Table B.2: Regressions of Turn-taking Attempts and Coordination Instability (T=4)

	Group Attempts to Take Turns			Coordination Instability		
	(1) Block1	(2) Block2	(3) Block3	(4) Block1	(5) Block2	(6) Block3
No ID	-0.018 (0.086)	0.002 (0.082)	0.014 (0.074)	0.005 (0.016)	0.015 (0.017)	0.009 (0.023)
Het	-0.028 (0.077)	-0.032 (0.092)	-0.051 (0.089)	0.079*** (0.022)	0.010 (0.013)	-0.004 (0.020)
4P	-0.215*** (0.078)	-0.401*** (0.101)	-0.469*** (0.100)	0.122*** (0.021)	0.096*** (0.023)	0.032 (0.027)
No ID $\times$ 4P	-0.063 (0.106)	0.068 (0.126)	0.107 (0.131)	-0.041 (0.025)	-0.033 (0.031)	0.014 (0.033)
Het $\times$ 4P	0.013 (0.097)	0.137 (0.128)	0.141 (0.138)	-0.065** (0.030)	0.005 (0.027)	0.009 (0.030)
Coordination Instability	-0.291** (0.115)	-0.421* (0.235)	-0.602*** (0.219)			
Group Attempts to Take Turns				-0.054** (0.023)	-0.035* (0.020)	-0.045** (0.018)
Constant	0.837*** (0.061)	0.886*** (0.063)	0.906*** (0.060)	0.103*** (0.024)	0.043* (0.023)	0.061** (0.028)
<i>Implied differences in large groups</i>						
4P, No ID - 4P, Hom	-0.081 (0.062)	0.070 (0.095)	0.121 (0.108)	-0.036* (0.020)	-0.018 (0.026)	0.023 (0.024)
4P, Het - 4P, Hom	-0.016 (0.060)	0.105 (0.089)	0.090 (0.106)	0.014 (0.020)	0.015 (0.023)	0.005 (0.022)
Observations	700	700	700	700	700	700

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns 1 to 6: Group-Window level OLS regressions with standard errors clustered at the group level in parentheses. Group Attempts to Take Turns is the average of individual attempts to take turns within a group in each window. Group Attempts to Take Turns and Coordination Instability are calculated for every 4 consecutive rounds. Since each block consists of 20 rounds, there are 5 observations for every group in each column.

C Mechanisms

To proxy for how willing participants are to take turns, we measure each individual i 's attempt to take turns over an even window of T rounds as:

$$\text{Turn-taking Attempt}_T^i = 1 - \frac{\left| \frac{\sum_{t \in T} A_t^i}{T} - 0.5 \right|}{0.5}$$

where A_t^i is a binary variable capturing whether i chose Market A in round t which belongs to the window of T rounds. Intuitively, an individual who is trying to take-turns perfectly would alternate between Market A and Market B, spending half their time in each. This measure has a minimum of 0 when an individual always chooses the same action, and a maximum of 1 when he/she alternates (perfectly) between the markets.²⁰

Each group g 's attempt to take turns over an even window of T rounds is just the average of the group members' individual measures:

$$\text{Turn-taking Attempt}_T^g = \frac{1}{N} \sum_{i \in g} \text{Turn-taking Attempt}_T^i$$

To proxy for the difficulty of sustaining coordination, we measure each group g 's coordination instability over an even window of T rounds as:

$$\text{Coordination Instability}_T^g = \sqrt{\frac{\sum_{t \in T} (P_t - \bar{P}_T)^2}{T - 1}}$$

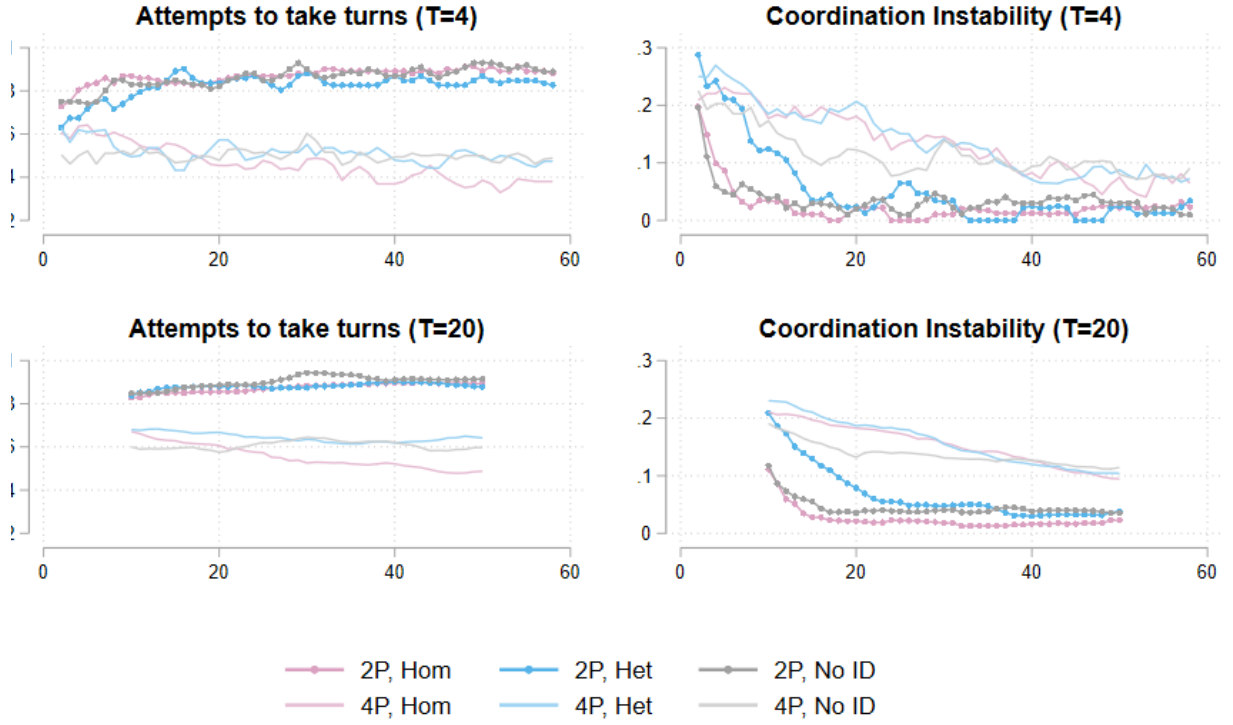
where P_t is the proportion of the group in Market A in round t and $\bar{P}_T$ is the mean of P_t across all rounds in the window. This is the sample standard deviation of the variable P_t in the window. Intuitively, groups who have already coordinated on an equilibrium should not have much variation in the proportion who choose Market A. A higher standard deviation suggests that conditional on their mean rate of choosing Market A, groups are still miscoordinating in terms of the timings of their choices.

Figure C.1 depicts the evolution of turn-taking attempts and coordination instability over time. Similar to the patterns observed for efficiency and equity, persistent gaps emerge between groups of different sizes, along with notable differences in how group composition affects behav-

²⁰The RHS term is an inverse measure of turn-taking attempts: its numerator measures deviations from this benchmark while its denominator normalizes it to be between 0 and 1. Subsequently, we invert this measure to get a measure of turn-taking attempts.

ior within small and large groups.

Figure C.1: Turn-taking Attempts and Coordination Instability over Treatments



[†] Values are computed using a window of size $T = 4$ and $T = 20$. For $T = 4$, the value at time t is calculated as the average of 4 consecutive rounds, $(t - 1, t, t + 1, t + 2)$. For $T = 20$, the moving average at time t covers 20 rounds, $(t - 9, \dots, t, \dots, t + 10)$. Observations without a full window are omitted.

D Cluster Analysis

The k-means clustering algorithm is a classical unsupervised learning which can automatically group similar observations into one class according to the characteristics of the data (Hartigan and Wong, 1979). As an unsupervised learning algorithm, one key feature is that it finds potential patterns through iterative calculations and does not need category labels for the data in advance. The algorithm aims to partition the given dataset into k clusters, ensuring that data points are more similar within each cluster and less similar between clusters.

To capture individuals' pattern of choices, we construct a vector of three different elements for each block of 20 rounds.²¹ In Block 1, these elements are calculated using rounds 2 to 20, excluding round 1 because there is no prior history.²²

- (1) The rate of choosing A across all rounds.
- (2) The rate of choosing A when choosing A in the previous round.
- (3) The rate of choosing A when choosing B in the previous round.

Element (1) captures overall, how evenly they switch between Market A and Market B. Elements (2) and (3) capture the extent to which their choices depend on past events. Theoretically, these variables can distinguish between several types of individual strategies: selfish exploiters, selfless compromisers, perfect turn-takers (who balance A and B every two rounds) and imperfect turn-takers (who alternate but not over two rounds).²³

We determine the optimal number of clusters using two commonly adopted methods in unsupervised learning: the Elbow Method and Silhouette Analysis. In our case, both methods point to a four-cluster solution as the most appropriate classification.

The Elbow Method evaluates the within-cluster sum of squares (WCSS) as the number of clusters increases. This inflection point (the "elbow") indicates diminishing returns in model fit from adding additional clusters. As shown in Figure D.1, panel (a), the WCSS sharply declines

²¹Some of these elements required some imputation. When no choice of A was ever made in the previous round, we assign an imputed value of 0 to element 2. When no choice of B was ever made in the previous round, we assign an imputed value of 1 to element 3.

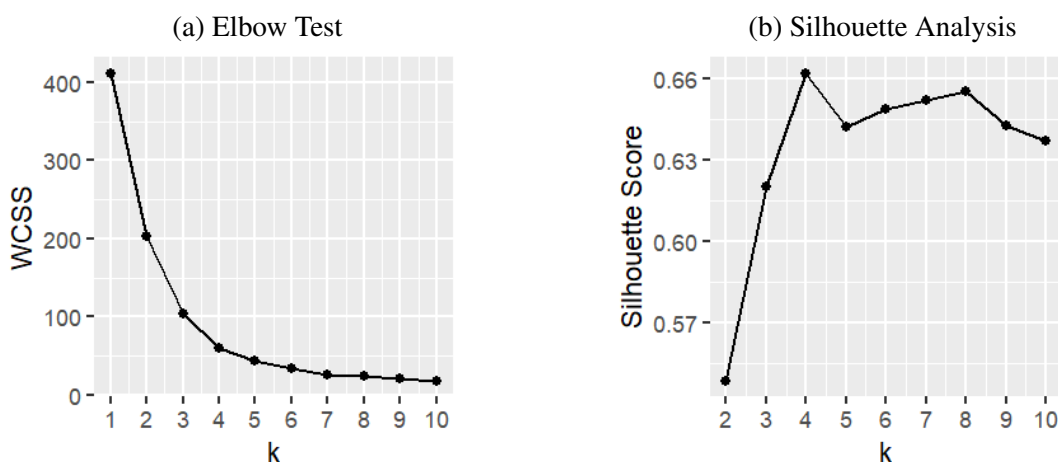
²²Using an odd window of 19 rounds in Block 1 rather than an even window does not affect the cluster analysis. When we instead use an even window of 18 rounds, from rounds 3 to 20, the resulting clusters are identical for 97.9% of observations, and all main results continue to hold. This additional analysis is available upon request.

²³A selfish individual who always wants the high payoff will have a vector of (1,1,1). A selfless, altruistic individual who accommodates others will have a vector of (0,0,0). A perfect turn-taker will have a vector of (0.5,1,0). An imperfect turn taker will have a vector of (0.5, X, Y), the values of X and Y are undetermined as long as the entry rate is around 0.5.

from $k = 2$ to $k = 4$, after which the rate of decrease flattens. According to the elbow method, $k = 4$ is considered to be a suitable clustering number.

The Silhouette Analysis, displayed in Figure D.1, panel (b), provides a complementary perspective by measuring how well each observation fits within its assigned cluster relative to other clusters. The silhouette coefficient ranges from -1 to 1, with higher values indicating better-defined clusters. The analysis shows that $k = 4$ yields the highest average silhouette score, suggesting that observations are most coherently grouped when four clusters are used.

Figure D.1: Tests to determine the number of k-means clusters



To ensure robustness and avoid local convergence of the k-means procedure, we initialized the algorithm 50 times and retained the solution with the lowest total within-cluster sum of squares. To assess the stability of the clustering solution, we replicated the k-means procedure 1,000 times with the same parameters.

The clustering procedure reveals that individual behavior across blocks can be categorized into four distinct types. Table D.1 shows the characteristics of each of them. Notice that an individual might be classified as different types across blocks if their strategies have evolved.

1. **Perfect Turn-takers** have a strategy that is very close to alternating between A and B every round. Consequently, they have an almost maximal level of turn-taking intentions. We call them *perfect turn-takers* to reflect this strategy. They play the equilibrium that allows for efficiency and equity.
2. **Imperfect Turn-takers** also switch their choices between two options, but not as frequently as perfect turn-takers. They tend to stay in one choice for more than one round

and make alternations in the long run. Consequently, they have a relatively high level of attempts to take turns. We call them *imperfect turn-takers* to reflect this strategy.

3. **Exploiters** choose A irrespective of their choice in past round. Consequently, they have a very low level of attempts to take turns. We call them *exploiters* to reflect their predominantly aggressive market entry strategy.
4. **Compromisers** tend to choose A only one-fifth of the time, with limited alternation between the options. Within this category, individuals typically opt to remain in the less profitable Market B for most rounds. They demonstrate compromise and inertia in decision-making, diverging from the alternation strategy. Consequently, they have few attempts to take turns. We call them *compromisers* to reflect how they give in to others by not choosing Market A.

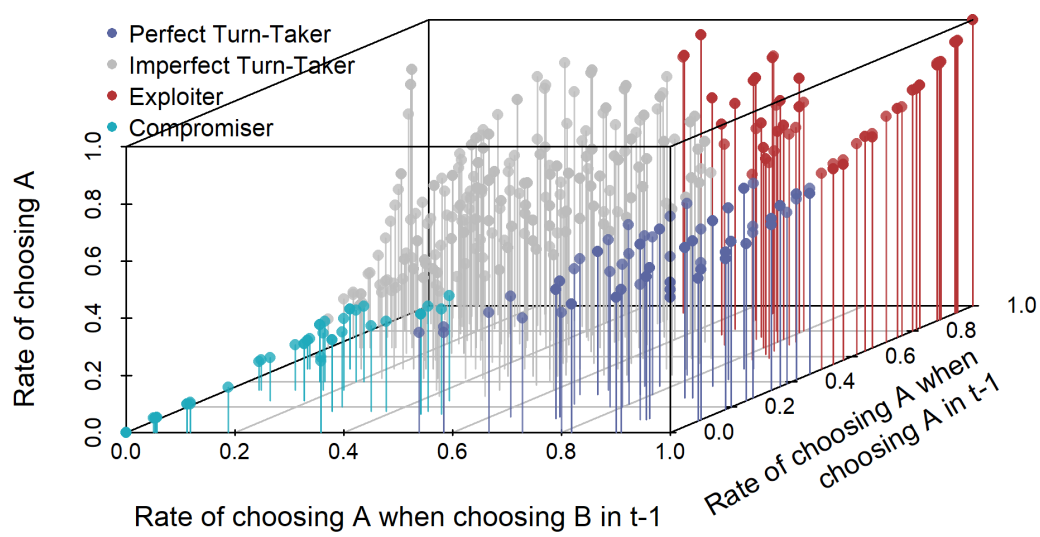
Table D.1: Results of Cluster Analysis

	(1) Element 1	(2) Element 2	(3) Element 3	(4) Attempts to Take Turns ($T = 20$)
Perfect Turn-Taker	0.508	0.050	0.975	0.973
Imperfect Turn-Taker	0.542	0.700	0.363	0.756
Exploiter	0.913	0.905	0.924	0.180
Compromiser	0.088	0.094	0.092	0.177

Uses individual-block level data for Elements 1 to 3 to perform the cluster analysis. Columns 1 to 3 show the means of each element for each cluster. Column 4 shows the mean of attempts to take turns within a block for individuals classified into that type.

In Figure D.2, we plot data points in different colors for each type assigned by the k-means procedure. Each point represents a participant's behavior in each of the three blocks.

Figure D.2: k-means clustering



E Conceptual Framework

In this section, we use a standard rational game theory framework to demonstrate how group sizes and identity could affect the complexity of coordination.

The market entry game is a typical repeated game, where each stage game has multiple Nash equilibria. In particular, focusing on symmetric Nash equilibria, the Nash equilibria can be classified into two types: (i) one with pure strategies, where half the group chooses Market A and the other half chooses Market B, and (ii) one where everyone plays the same mixed strategy.

Given the multiplicity of such Nash equilibria, there are also numerous Subgame Perfect Nash equilibria – with both history-independent and history-dependent strategies – where both efficiency and equity across rounds are (eventually) reached. However, it is unlikely that history-independent equilibria with both efficiency and equity will emerge spontaneously without any prior communication.²⁴ More likely, players would adjust their strategies after observing the actions of others, finally reaching some turn-taking arrangement.

A simple SPNE which has such characteristics is the turn-taking with independent randomisations equilibria highlighted in [Lau and Mui \(2008, 2012\)](#). We describe an equilibrium with such characteristics in the following proposition:

Proposition 1. *The following strategies constitute an SPNE: In the initial round, play the symmetric mixed strategy NE. In subsequent rounds, if there is an outcome where half the group chooses Market A, play the opposite strategy from what one chose in the previous round. Otherwise, continue playing the symmetric mixed strategy NE.*

Intuition: In this SPNE, players initially randomise their choices of Market A and Market B which captures some trial and error in players trying to reach a turn-taking arrangement. This randomisation leads to both non-efficient and efficient outcomes. When observing non-efficient outcomes like (3A, 1B) in large groups or (2A) in small groups, it is still not clear who should do what in the next round to achieve efficiency and equity. However, when observing efficient outcomes like (2A, 2B) or (1A, 1B) in small groups, alternating one’s strategy will create efficiency and equity henceforth. This thus serves as a viable “signal” for them to begin turn-taking.

Given the relatively intuitive nature of this SPNE, we focus on this as our theoretical benchmark. For this SPNE, we can use the mixed strategy NE strategies to calculate the chance of

²⁴An example of a history-independent strategy with both efficiency and equity is one where half choose Market A in odd rounds, and the other half choose Market A in even rounds. Here, it would be unclear who this “half” refers to.

reaching a turn-taking arrangements. It turns out that turn-taking is easier in smaller groups than larger groups as expected:

Corollary 1 (Effects of group size). *In the above SPNE, the probability of reaching a turn-taking arrangement in any (initial) round is higher in small groups ($p = 0.42$) than in larger groups ($p = 0.321$).*

Proof. In small (large) groups, the symmetric mixed strategy stage NE is to choose Market A with probability 0.7 (0.63674), which gives a chance of reaching this in each round with probability $0.7 \times 0.6 \times 2 = 0.42$. $(0.63674^2 \times (1 - 0.63674)^2 \times 4C2 = 0.321)$ $\square$

When players possess identities, this will not make a difference compared to when there are no identities in situations where a group is homogeneous – i.e. all players in the group possess the same identities. However, when players in a group possess different identities (in equal proportions), such identities also create salient markers which potentially help improve on reaching an equitable and efficient turn-taking arrangement. The following proposition demonstrates an example of an SPNE where this is achieved:

Proposition 2. *The following strategies constitute an SPNE: In the initial round, play the symmetric mixed strategy NE. In subsequent rounds, (i) if there is an outcome in the past where half the group chooses Market A, then play the opposite strategy from what one chose in the previous round. (ii) If there is an outcome in the past where one's (the other's) identity group chooses the same market and the other (one's) identity group does not do the same, then play the opposite strategy from what one's identity group chose in the previous round (play the same strategy as what one's identity group chose in the previous round). Otherwise, continue playing the symmetric mixed strategy NE.*

Intuition: This SPNE follows the same structure as before where a certain event triggers turn-taking. However, in addition to the case where one observes an efficient outcome, in heterogeneous groups, there is also a case where one identity group, or the other identity group managed to coordinate on their actions. If so, then they will also be able to begin a turn-taking arrangement.

Notice that the latter case only differs from the former when the group size is larger ($N = 4$) and not when it is small ($N = 2$). For example, consider a case where, in the previous round, both Blue members chose Market A, while one Red member chose Market A and the other chose Market B. Then, in the current round, they will begin a turn-taking arrangement, starting with Blue members choosing Market B and Red members choosing Market A. Such cases do not

exist for $N = 2$, and reaching a turn-taking arrangement is only possible because of the distinct identity labels present.

Corollary 2 (Effects of identities). *There is no difference in the probability of reaching a turn-taking arrangement in groups which are homogeneous or have no identities. In the above SPNE, compared to such groups, in heterogeneous groups, the probability of reaching a turn-taking arrangement in any (initial) round is higher in larger groups but has no difference in two-player groups.*

While the SPNE discussed are very specific, we believe that they illustrate how group sizes and identities could affect turn-taking arrangements. A caveat is that this would likely only hold in a rational framework. Most certainly, other factors like bounded rationality and other-regarding preferences may also affect the extent to which group sizes and identities work.

F Experiment Materials

F.1 Instructions

Treatment: Small, Homogeneous and Small, Heterogeneous

General Information

Welcome to the experiment.

You are now taking part in an interactive study concerned on decision making. **Please pay attention to the information provided here and make your decisions carefully. If at any time you have questions to ask, please raise your hand and we will attend to you in private.**

Do note that unauthorized communication is prohibited. Failure to adhere to this rule would force us to stop the study and you may be held liable for the costs incurred. *Do not write anything on this instruction sheet.*

Your anonymity will be preserved for the study. You and other participants will never be aware of the personal identities of other participants during or after the study. Participants will only be identified by a computer-generated numerical ID in our data collection. All information collected will strictly be kept confidential for the sole purpose of this study.

By participating in this study, you will be able to earn an amount of money. The amount depends on the decisions you and others make. All earnings will be denominated in Experimental Currency Units (ECUs). The conversion rate is 1000 ECUs = 1 SGD. In addition, you will also receive a showup fee of 2 SGD. At the end of this session, this money will be paid to you privately and confidentially.

In this experiment, there will be a total of 3 stages. Below are the instructions for Stage 1. You will receive the instructions for Stages 2 and 3 after Stage 1 is completed.

Stage 1

In Stage 1, you will firstly be randomly assigned by the computer to either a **BLUE** group, or a **RED** group. The group which you have been assigned will remain the same *throughout the experiment*.

Spot-the Differences Task

You will then have to complete a spot-the differences task together with other members of your group. There will be a total of 2 puzzles.

In each puzzle, your group will be shown a pair of pictures and have 3 minutes to find the differences in each puzzle. One member from your group will be randomly appointed as the representative to submit the answers on your behalf.

You will be given the opportunity to chat with your group members to find the differences. You are encouraged to actively participate in this chat by contributing your suggested answers and helping your group to solve these puzzles.

Payoffs

For each difference found by the group, you will earn 200 ECUs. You will learn about your group's results at the end of this stage.

Stage 2

In Stage 2, you will be randomly matched with another participant and will complete several rounds of a decision task with them; more details are provided below. Given the random matching, the other participant may or may not belong to the same group (**BLUE** or **RED**) as you. You will be notified of their groups at the beginning of this stage. Note that this matching will remain the same for the rest of the experiment.

The Decision Task

In this task, *you and the other participant* will have to choose an activity to join over 60 rounds.

In each round, you and the other participants will each decide which activity, A or B, to join. Your choice, as well as the choices made by the other participant, will determine your payoffs for that round.

Payoffs

Your payoffs will vary according to which activity you choose and the total number of other participants in that activity as described in the *payoff table* below.

Number who join Activity A	1	2
Payoff if in Activity A	400	50
Number who join Activity B	1	2
Payoff if in Activity B	200	50

For example, suppose 1 participant joins Activity A, while the other participant joins Activity B. Then the participant who joins Activity A will receive 400 ECUs while the participant who joins activity B will receive 200 ECUs.

Information provided

In each round, you will be able to see the following to aid in your decision.

1. The relevant payoff table.
2. A record of past decisions (which activity joined) made by you and the other participant.

Results

At the end of each round, after you and the other participant have submitted your decisions, you will see the choices of everyone in the round and your individual payoffs for the round in ECUs.

Total Payment

Your total payoffs are equal to the sum of earnings across all rounds in the decision task.

Stage 3

In Stage 3, you will have to answer a series of survey questions and complete some further tasks.

Treatment: Small, No identity

General Information

Welcome to the experiment.

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By participating in this study, you will be able to earn an amount of money. The amount depends on the decisions you and others make. All earnings will be denominated in Experimental Currency Units (ECUs). The conversion rate is 1000 ECUs = 1 SGD. In addition, you will also receive a showup fee of 2 SGD. At the end of this session, this money will be paid to you privately and confidentially.

In this experiment, there will be a total of 3 stages. Below are the instructions for Stage 1. You will receive the instructions for Stages 2 and 3 after Stage 1 is completed.

Stage 1

Spot-the Differences Task

In Stage 1, you will have to complete a spot-the differences task together with a group of other participants. There will be a total of 2 puzzles.

In each puzzle, your group will be shown a pair of pictures and have 3 minutes to find the differences in each puzzle. One member from your group will be randomly appointed as the representative to submit the answers on your behalf.

You will be given the opportunity to chat with your group members to find the differences. You are encouraged to actively participate in this chat by contributing your suggested answers and helping your group to solve these puzzles.

Payoffs

For each difference found by the group, you will earn 200 ECUs. You will learn about your group's results at the end of this stage.

Stage 2

In Stage 2, you will be randomly matched with another participant and will complete several rounds of a decision task with them; more details are provided below. Note that this matching will remain the same for the rest of the experiment.

The Decision Task

In this task, *you and the other participant* will have to choose an activity to join over 60 rounds.

In each round, you and the other participants will each decide which activity, A or B, to join. Your choice, as well as the choices made by the other participant, will determine your payoffs for that round.

Payoffs

Your payoffs will vary according to which activity you choose and the total number of other participants in that activity as described in the *payoff table* below.

Number who join Activity A	1	2
Payoff if in Activity A	400	50
Number who join Activity B	1	2
Payoff if in Activity B	200	50

For example, suppose 1 participant joins Activity A, while the other participant joins Activity B. Then the participant who joins Activity A will receive 400 ECUs while the participant who joins activity B will receive 200 ECUs.

Information provided

In each round, you will be able to see the following to aid in your decision.

1. The relevant payoff table.
2. A record of past decisions (which activity joined) made by you and the other participant.

Results

At the end of each round, after you and the other participant have submitted your decisions, you will see the choices of everyone in the round and your individual payoffs for the round in ECUs.

Total Payment

Your total payoffs are equal to the sum of earnings across all rounds in the decision task.

Stage 3

In Stage 3, you will have to answer a series of survey questions and complete some further tasks.

Treatment: Large, Homogeneous and Large, Heterogeneous

General Information

Welcome to the experiment.

You are now taking part in an interactive study concerned on decision making. **Please pay attention to the information provided here and make your decisions carefully. If at any time you have questions to ask, please raise your hand and we will attend to you in private.**

Do note that unauthorized communication is prohibited. Failure to adhere to this rule would force us to stop the study and you may be held liable for the costs incurred. *Do not write anything on this instruction sheet.*

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By participating in this study, you will be able to earn an amount of money. The amount depends on the decisions you and others make. All earnings will be denominated in Experimental Currency Units (ECUs). The conversion rate is 1000 ECUs = 1 SGD. In addition, you will also receive a showup fee of 2 SGD. At the end of this session, this money will be paid to you privately and confidentially.

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You will be given the opportunity to chat with your group members to find the differences. You are encouraged to actively participate in this chat by contributing your suggested answers and helping your group to solve these puzzles.

Payoffs

For each difference found by the group, you will earn 200 ECUs. You will learn about your group's results at the end of this stage.

Stage 2

In Stage 2, you will be randomly matched with 3 other participants and will complete several rounds of a decision task with them; more details are provided below. Given the random matching, the 3 other participants may or may not belong to the same group (**BLUE** or **RED**) as you. You will be notified of their groups at the beginning of this stage. Note that this matching will remain the same for the rest of the experiment.

The Decision Task

In this task, *you and the other 3 participants* will have to choose an activity to join over 60 rounds.

In each round, you and the other 3 participants will each decide which activity, A or B, to join. Your choice, as well as the choices made by the 3 others, will determine your payoffs for that round.

Payoffs

Your payoffs will vary according to which activity you choose and the total number of other participants in that activity as described in the *payoff table* below.

Number who join Activity A	1	2	3	4
Payoff if in Activity A	400	400	50	50
Number who join Activity B	1	2	3	4
Payoff if in Activity B	200	200	50	50

For example, suppose 2 participants join Activity A, while the other 2 participants join Activity B. Then each of those who join Activity A will receive 400 ECUs while each of those who join activity B will receive 200 ECUs.

Information provided

In each round, you will be able to see the following to aid in your decision.

1. The relevant payoff table.
2. A record of past decisions (which activity joined) made by you and the 3 other participants.

Results

At the end of each round, after you and the other 3 participants have submitted your decisions, you will see the choices of everyone in the round and your individual payoffs for the round in ECUs.

Total Payment

Your total payoffs are equal to the sum of earnings across all rounds in the decision task.

Stage 3

In Stage 3, you will have to answer a series of survey questions and complete some further tasks.

Treatment: Large, No identity

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The Decision Task

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In each round, you and the other 3 participants will each decide which activity, A or B, to join. Your choice, as well as the choices made by the 3 others, will determine your payoffs for that round.

Payoffs

Your payoffs will vary according to which activity you choose and the total number of other participants in that activity as described in the *payoff table* below.

Number who join Activity A	1	2	3	4
Payoff if in Activity A	400	400	50	50
Number who join Activity B	1	2	3	4
Payoff if in Activity B	200	200	50	50

For example, suppose 2 participants join Activity A, while the other 2 participants join Activity B. Then each of those who join Activity A will receive 400 ECUs while each of those who join activity B will receive 200 ECUs.

Information provided

In each round, you will be able to see the following to aid in your decision.

1. The relevant payoff table.
2. A record of past decisions (which activity joined) made by you and the 3 other participants.

Results

At the end of each round, after you and the other 3 participants have submitted your decisions, you will see the choices of everyone in the round and your individual payoffs for the round in ECUs.

Total Payment

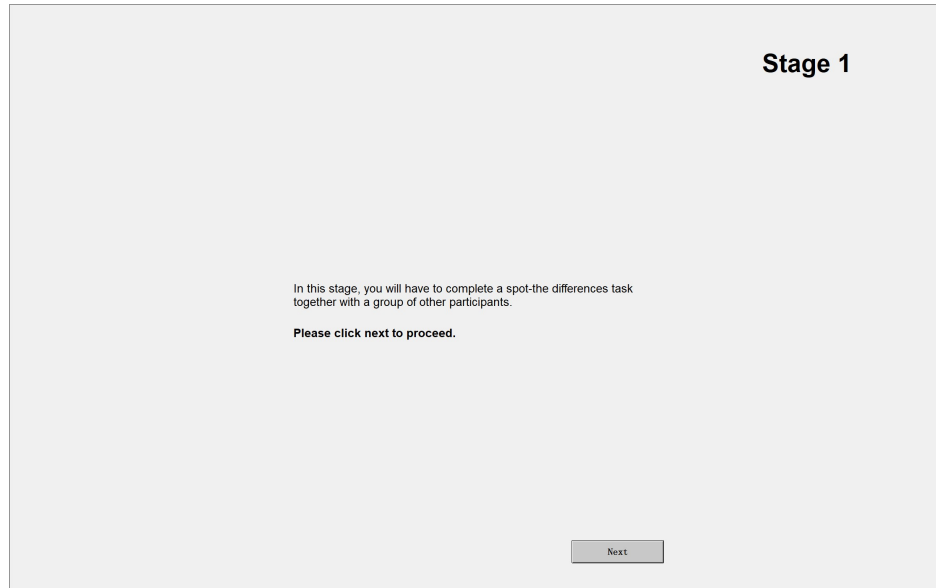
Your total payoffs are equal to the sum of earnings across all rounds in the decision task.

Stage 3

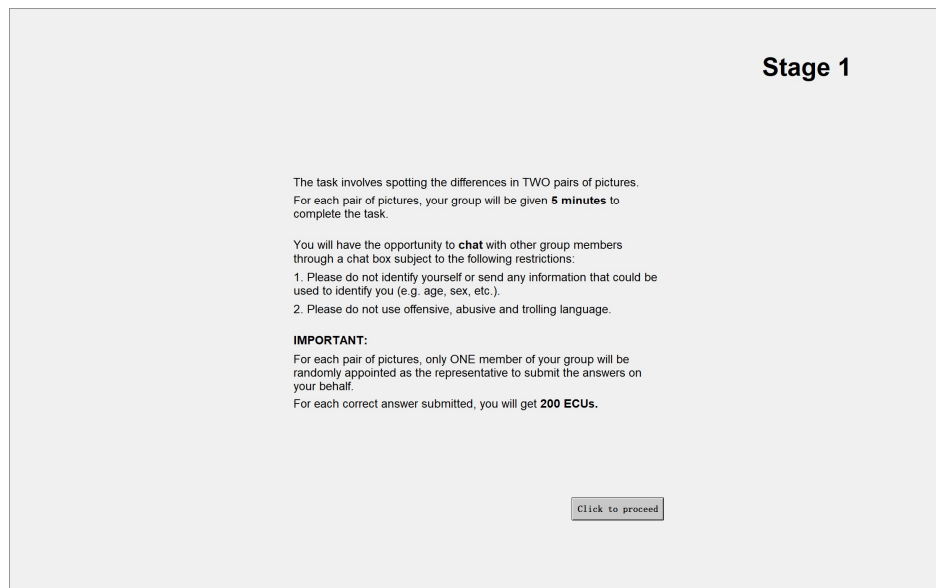
In Stage 3, you will have to answer a series of survey questions and complete some further tasks.

F.2 Screenshots

Stage 1 (No Identity)



Introduction



Joint Task: Instructions

Please find the five differences in the pair of pictures.
You are encouraged to actively discuss and help your group to solve the puzzle.

Time Left[sec]: 288

Puzzle 1

	1	2	3	4	5	6
A						
B						
C						
D						
E						
F						

The member of your group with **ID 3** has been chosen to submit your group answers on behalf of all members of your group .

ID 3 can select the answers at any time, but can only submit them in the last 30 seconds.

Submission of answers will open in the last 30 seconds.

Type a message and press enter to send.
Your (randomly generated) chat ID is **ID 2**.

Joint Task: Puzzle 1 (Non-Submitter)

Please find the five differences in the pair of pictures.
You are encouraged to actively discuss and help your group to solve the puzzle.

Time Left[sec]: 157

Puzzle 1

	1	2	3	4	5	6
A						
B						
C						
D						
E						
F						

You have been chosen to submit the answers on behalf of all members of your group.
Please be mindful of the time limit. The order of your answers does NOT matter.
You can select the answers at any time, but can only submit them in the last 30 seconds.

1st Difference	☐ A ☐ B ☐ C ☐ D ☐ E ☐ F	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	2nd Difference	☐ A ☐ B ☐ C ☐ D ☐ E ☐ F	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
3rd Difference	☐ A ☐ B ☐ C ☐ D ☐ E ☐ F	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	4th Difference	☐ A ☐ B ☐ C ☐ D ☐ E ☐ F	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
5th Difference	☐ A ☐ B ☐ C ☐ D ☐ E ☐ F	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6			

Type a message and press enter to send.
Your (randomly generated) chat ID is **ID 3**.

Joint Task: Puzzle 1 (Submitter)

Please find the six differences in the pair of pictures.
You are encouraged to actively discuss and help your group to solve the puzzle.

Time Left[sec]:277

Puzzle 2

	1	2	3	4	5	6
A						
B						
C						
D						
E						

The member of your group with **ID 4** has been chosen to submit your group answers on behalf of all members of your group .

ID 4 can select the answers at any time, but can only submit them in the last 30 seconds.

Submission of answers will open in the last 30 seconds.

Type a message and press enter to send.
Your (randomly generated) chat ID is ID 3.

Joint Task: Puzzle 2 (Non-Submitter)

Please find the six differences in the pair of pictures.
You are encouraged to actively discuss and help your group to solve the puzzle.

Time Left[sec]:255

Puzzle 2

	1	2	3	4	5	6
A						
B						
C						
D						
E						

You have been chosen to submit the answers on behalf of all members of your group.
Please be mindful of the time limit. The order of your answers does NOT matter.
You can select the answers at any time, but can only submit them in the last 30 seconds.

1st Difference ☐ A ☐ B ☐ C ☐ D ☐ E	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	2nd Difference ☐ A ☐ B ☐ C ☐ D ☐ E	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
3rd Difference ☐ A ☐ B ☐ C ☐ D ☐ E	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	4th Difference ☐ A ☐ B ☐ C ☐ D ☐ E	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
5th Difference ☐ A ☐ B ☐ C ☐ D ☐ E	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	6th Difference ☐ A ☐ B ☐ C ☐ D ☐ E	☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6

Type a message and press enter to send.
Your (randomly generated) chat ID is ID 4.

Joint Task: Puzzle 2 (Submitter)

Puzzle	Your Group's Answers					The Correct Answers					
1	N. A.	N. A.	N. A.	N. A.	N. A.	A5	B1	D2	D6	F1	
2	N. A.	N. A.	N. A.	N. A.	N. A.	N. A.	A6	B1	C3	D3	D4 E4
Your group correctly pointed out 0 out of 11 differences! Your payoff in this stage is 0 ECUs. Thank you for spending time working with your peers from your group ! Please click the OK button to enter Stage 2 of the Experiment!											
											OK

Joint Task: Results

Stage 1 (With Identity)

Stage 1

You have been assigned to:

Group BLUE

Note that the group which you have been assigned will remain the same throughout the experiment.

In this stage, you will work together with other members of Group **BLUE** to complete a Group Task.

Please click next to proceed.

Next

Introduction

Stage 1

The task involves spotting the differences in TWO pairs of pictures.
For each pair of pictures, your group will be given **5 minutes** to complete the task.

You will have the opportunity to **chat** with other group members through a chat box subject to the following restrictions:

1. Please do not identify yourself or send any information that could be used to identify you (e.g. age, sex, etc.).
2. Please do not use offensive, abusive and trolling language.

IMPORTANT:
For each pair of pictures, only ONE member of your group will be randomly appointed as the representative to submit the answers on your behalf.
For each correct answer submitted, you will get **200 ECUs**.

Click to proceed

Joint Task: Instructions

Please find the five differences in the pair of pictures.
You are encouraged to actively discuss and help your Group **BLUE** to solve the puzzle.

Puzzle 1

	1	2	3	4	5	6
A						
B						
C						
D						
E						
F						

Time Left[sec]: 274

Participants you are working with:

you

The member of your group with **Group ID 3** has been chosen to submit your group answers on behalf of all members of your Group **BLUE**.

Group ID 3 can select the answers at any time, but can only submit them in the last 30 seconds.

Submission of answers will open in the last 30 seconds.

Type a message and press enter to send.

Your (randomly generated) chat ID is **Group ID 4**.

Joint Task: Puzzle 1 (Non-Submitter)

Please find the five differences in the pair of pictures.
You are encouraged to actively discuss and help your Group **BLUE** to solve the puzzle.

Puzzle 1

	1	2	3	4	5	6
A						
B						
C						
D						
E						
F						

Time Left[sec]: 242

Participants you are working with:

you

You have been chosen to submit the answers on behalf of all members of your Group **BLUE**.

Please be mindful of the time limit. The order of your answers does NOT matter.

You can select the answers at any time, but can only submit them in the last 30 seconds.

1st Difference <table> <tr><td>☐ A</td><td>☐ 1</td></tr> <tr><td>☐ B</td><td>☐ 2</td></tr> <tr><td>☐ C</td><td>☐ 3</td></tr> <tr><td>☐ D</td><td>☐ 4</td></tr> <tr><td>☐ E</td><td>☐ 5</td></tr> <tr><td>☐ F</td><td>☐ 6</td></tr> </table>	☐ A	☐ 1	☐ B	☐ 2	☐ C	☐ 3	☐ D	☐ 4	☐ E	☐ 5	☐ F	☐ 6	2nd Difference <table> <tr><td>☐ A</td><td>☐ 1</td></tr> <tr><td>☐ B</td><td>☐ 2</td></tr> <tr><td>☐ C</td><td>☐ 3</td></tr> <tr><td>☐ D</td><td>☐ 4</td></tr> <tr><td>☐ E</td><td>☐ 5</td></tr> <tr><td>☐ F</td><td>☐ 6</td></tr> </table>	☐ A	☐ 1	☐ B	☐ 2	☐ C	☐ 3	☐ D	☐ 4	☐ E	☐ 5	☐ F	☐ 6
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☐ B	☐ 2																								
☐ C	☐ 3																								
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☐ A	☐ 1																								
☐ B	☐ 2																								
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5th Difference <table> <tr><td>☐ A</td><td>☐ 1</td></tr> <tr><td>☐ B</td><td>☐ 2</td></tr> <tr><td>☐ C</td><td>☐ 3</td></tr> <tr><td>☐ D</td><td>☐ 4</td></tr> <tr><td>☐ E</td><td>☐ 5</td></tr> <tr><td>☐ F</td><td>☐ 6</td></tr> </table>	☐ A	☐ 1	☐ B	☐ 2	☐ C	☐ 3	☐ D	☐ 4	☐ E	☐ 5	☐ F	☐ 6													
☐ A	☐ 1																								
☐ B	☐ 2																								
☐ C	☐ 3																								
☐ D	☐ 4																								
☐ E	☐ 5																								
☐ F	☐ 6																								

Type a message and press enter to send.

Your (randomly generated) chat ID is **Group ID 3**.

Joint Task: Puzzle 1 (Submitter)

Please find the six differences in the pair of pictures.
You are encouraged to actively discuss and help your Group **BLUE** to solve the puzzle.

Puzzle 2

	1	2	3	4	5	6
A						
B						
C						
D						
E						

Group ID 3: Hi!!!

Type a message and press enter to send.
Your (randomly generated) chat ID is **Group ID 3**.

Time Left[sec]: 231

Participants you are working with:

you

The member of your group with **Group ID 4** has been chosen to submit your group answers on behalf of all members of your Group **BLUE**.

Group ID 4 can select the answers at any time, but can only submit them in the last 30 seconds.

Submission of answers will open in the last 30 seconds.

Joint Task: Puzzle 2 (Non-Submitter)

Please find the six differences in the pair of pictures.
You are encouraged to actively discuss and help your Group **BLUE** to solve the puzzle.

Puzzle 2

	1	2	3	4	5	6
A						
B						
C						
D						
E						

Time Left[sec]: 267

Participants you are working with:

YOU

You have been chosen to submit the answers on behalf of all members of your Group **BLUE**.
Please be mindful of the time limit. The order of your answers does NOT matter.
You can select the answers at any time, but can only submit them in the last 30 seconds.

1st Difference ☐ A ☐ B ☐ C ☐ D ☐ E ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	2nd Difference ☐ A ☐ B ☐ C ☐ D ☐ E ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
3rd Difference ☐ A ☐ B ☐ C ☐ D ☐ E ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	4th Difference ☐ A ☐ B ☐ C ☐ D ☐ E ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6
5th Difference ☐ A ☐ B ☐ C ☐ D ☐ E ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6	6th Difference ☐ A ☐ B ☐ C ☐ D ☐ E ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6

Type a message and press enter to send.
Your (randomly generated) chat ID is **Group ID 4**.

Joint Task: Puzzle 2 (Submitter)

Participants you are working with:

you

Puzzle	Your Group's Answers	The Correct Answers
1	N. A. N. A. N. A. N. A. N. A.	A5 B1 D2 D6 F1
2	N. A. N. A. N. A. N. A. N. A. N. A.	A6 B1 C3 D3 D4 E4

Your group correctly pointed out 0 out of 11 differences!
Your payoff in this stage is 0 ECUs.
Thank you for spending time working with your peers from your Group BLUE !
Please click the OK button to enter Stage 2 of the Experiment!

OK

Joint Task: Results

Stage 2 (All)

Quiz

You will now have a short quiz to test your understanding of the decision task in Stage 2.

Do answer all of the questions, you will have to answer all of them correctly to proceed.

Raise your hand if you are having trouble answering any of the questions.

Please click next to proceed.

Next

Understanding Test: Instructions

Stage 2: Understanding Test (Large Groups)

Quiz

Q1. In Stage 2, you will first be randomly matched with three other participants and then you will play a decision task for multiple periods.

Will the other participants you are matched with change across periods?

☐ Yes
☒ No

Your answer is correct, click next to proceed.

Next

Understanding Test Q1

Quiz

Number who join Activity A	1	2	3	4
Payoff if in Activity A	400	400	50	50
Number who join Activity B	1	2	3	4
Payoff if in Activity B	200	200	50	50

Q2. In each of the 5 cases below, what is the individual payoff of i) the participant(s) who joined Activity A and/or ii) the participant(s) who joined Activity B.

Cases a) to e)	Joined Activity A	Joined Activity B
a) 1 participant joined Activity A, 3 participants joined Activity B:	400	50
b) 2 participants joined Activity A, 2 participants joined Activity B:	400	200
c) 3 participants joined Activity A, 1 participant joined Activity B:	50	200
d) 4 participants joined Activity A:	50	
e) 4 participants joined Activity B:		50

Submit

Understanding Test Q2

Stage 2: Understanding Test (Small Groups)

Quiz

Q1. In Stage 2, you will first be randomly matched with another participants and then you will play a decision task for multiple periods.
Will the other participant you are matched with change across periods?

☐ Yes
☒ No

Your answer is correct, click next to proceed.

Next

Understanding Test Q1

Quiz

Number who join Activity A	1	2
Payoff if in Activity A	400	50
Number who join Activity B	1	2
Payoff if in Activity B	200	50

Q2. In each of the 3 cases below, what is the individual payoff of i) the participant(s) who joined Activity A and/or ii) the participant(s) who joined Activity B.

Cases a) to e)	Joined Activity A	Joined Activity B
a) 1 participant joined Activity A, 1 participant joined Activity B:	400	200
b) 2 participants joined Activity A:	50	
c) 2 participants joined Activity B:		50

Your answers are correct, click next to proceed.

Next

Understanding Test Q2

Stage 2: Decision Task (Small groups, with Identity)

Stage 2 Decision Task

Welcome to Stage 2 of the experiment!

This is a reminder that you belong to: Group **RED**

You have been randomly matched with another participant P1 .
Your assigned ID in this stage is P0.



P1



P2

This matching and assigned IDs will remain the same throughout Stage 2.

In each round, both of you will have to decide which activity to join.

Please click next to proceed.

Next

Decision Task Introduction

Round 1 Decision

Decision History



Round	P1	P2 (You)
1		

Payoff Table

Number who join Activity A	1	2
Payoff if in Activity A	400	50
Number who join Activity B	1	2
Payoff if in Activity B	200	50

Which activity would you like to join in this round?

☐ Activity A
☐ Activity B

Submit

Decision Task: Page 1

Decision History




Round	P1	P2 (You)
1	A	A

Payoff Table

Number who join Activity A	1	2
Payoff if in Activity A	400	50
Number who join Activity B	1	2
Payoff if in Activity B	200	50

Round 1 Result

In this round, **2 participant(s)** chose to join Activity **A**
and **0 participant(s)** chose to join Activity **B**.

You chose to join: **Activity A**

Your payoff in this round: **50** ECUs
Your total payoff: **50** ECUs

Please click next to proceed.

Next

Decision Task: Page 2

Stage 2: Decision Task (Small groups, no Identity)

**Stage 2
Decision Task**

Welcome to Stage 2 of the experiment!

You have been randomly matched with another participant P2 .
Your assigned ID is P1.
This matching and assigned IDs will remain the same throughout Stage 2.
In each round, both of you will have to decide which activity to join.

Please click next to proceed.

Next

Decision Task Introduction

**Round 1
Decision**

Decision History

Round	P1 (You)	P2
1		

Payoff Table

Number who join Activity A	1	2
Payoff if in Activity A	400	50
Number who join Activity B	1	2
Payoff if in Activity B	200	50

Which activity would you like to join in this round?

☐ Activity A
☐ Activity B

Submit

Decision Task: Page 1

Round 1 Result

Decision History

Round	P1	P2 (You)
1	A	A

Payoff Table

Number who join Activity A	1	2
Payoff if in Activity A	400	50
Number who join Activity B	1	2
Payoff if in Activity B	200	50

In this round, **2 participant(s)** chose to join Activity **A**
and **0 participant(s)** chose to join Activity **B**.

You chose to join: **Activity A**

Your payoff in this round: **50** ECUs

Your total payoff: **50** ECUs

Please click next to proceed.

Next

Decision Task: Page 2

Stage 2: Decision Task (Large groups, with Identity)

**Stage 2
Decision Task**

Welcome to Stage 2 of the experiment!

This is a reminder that you belong to Group **BI UF**

You have been randomly matched with 3 other participants P1, P2 & P4.
Your assigned ID in this stage is P3.


P1


P2


P3 (You)


P4

This matching and assigned IDs will remain the same throughout Stage 2.

In each round, the four of you will have to decide which activity to join.

Please click next to proceed.

Next

Decision Task Introduction

**Round 1
Decision**

Decision History



Round	P1	P2	P3 (You)	P4
1				

Payoff Table

Number who join Activity A	1	2	3	4
Payoff if in Activity A	400	400	50	50
Number who join Activity B	1	2	3	4
Payoff if in Activity B	200	200	50	50

Which activity would you like to join in this round?

☐ Activity A
☐ Activity B

Submit

Decision Task: Page 1

Round 1
Result

Decision History

				
Round	P1	P2	P3	P4 (You)
1	A	B	A	B

Payoff Table

Number who join Activity A	1	2	3	4
Payoff if in Activity A	400	400	50	50
Number who join Activity B	1	2	3	4
Payoff if in Activity B	200	200	50	50

In this round, **2 participant(s)** chose to join Activity **A**
and **2 participant(s)** chose to join Activity **B** .

You chose to join: **Activity B**

Your payoff in this round: **200** ECUs

Your total payoff: **200** ECUs

Please click next to proceed.

Next

Stage 2: Decision Task (Large groups, no Identity)

**Stage 2
Decision Task**

Welcome to Stage 2 of the experiment!

You have been randomly matched with 3 other participants P1, P2 & P3
Your assigned ID is P4.
This matching and assigned IDs will remain the same throughout Stage 2.
In each round, the four of you will have to decide which activity to join.

Please click next to proceed.

Next

Decision Task Introduction

**Round 1
Decision**

Decision History

Round	P1	P2	P3	P4 (You)
1				

Payoff Table

Number who join Activity A	1	2	3	4
Payoff if in Activity A	400	400	50	50
Number who join Activity B	1	2	3	4
Payoff if in Activity B	200	200	50	50

Which activity would you like to join in this round?

☐ Activity A
☐ Activity B

Submit

Decision Task: Page 1

Round 1 Result

Decision History

Round	P1	P2	P3 (You)	P4
1	A	A	A	A

Payoff Table

Number who join Activity A	1	2	3	4
Payoff if in Activity A	400	400	50	50
Number who join Activity B	1	2	3	4
Payoff if in Activity B	200	200	50	50

In this round, **4 participant(s)** chose to join Activity **A**
and **0 participant(s)** chose to join Activity **B**.

You chose to join: **Activity A**

Your payoff in this round: **50 ECUs**

Your total payoff: **50 ECUs**

Please click next to proceed.

Next

Decision Task: Page 2

Stage 3 (All)

**Decision Task
End**

We have reached the end of Stage 2, the Decision Task.

Your total payoff in this stage is 50 ECUs.

We will now proceed to Stage 3.

Please click next to proceed.

Next

End of Stage 2

Stage 3

Welcome to Stage 3 of the experiment!

You will now have to complete a series of short tasks and survey in the following pages.

Please click next to proceed.

Next

Stage 3: instructions

Stage 3: Allocation Task (with Identity)

Allocation Task

You will now play an allocation task with other participants.

In this allocation task, you will be given an endowment of 1000 ECUs and you will have to decide how much of this to allocate between two randomly chosen participants.

When making your decision, you will be given information about which group (BLUE or RED) the participants belong to.

As an allocatee, you will be randomly matched with another participant twice, with your allocation in each of these matches determined by a different randomly assigned allocator.

Note that as an allocator, the two allocatees you are matched with will never be responsible for allocating for you.

Please click next to proceed.

Allocation Task: Instructions

Allocation Task

You have been assigned to:
Group RED

As an allocator, please decide on your allocations in the following 3 scenarios.

1. Suppose you are allocating the 1000 ECUs between two participants from Group RED:


receives:

← 1000 ECUs →


receives:

2. Suppose you are allocating the 1000 ECUs between two participants from Group BLUE:


receives:

← 1000 ECUs →


receives:

3. Suppose you are allocating the 1000 ECUs between two participants with one of them from Group RED and another from Group BLUE:


receives:

← 1000 ECUs →


receives:

Allocation Task

Stage 3: Allocation Task (no Identity)

Allocation Task

You will now play an allocation task with other participants.

In this allocation task, you will be given an endowment of 1000 ECUs and you will have to decide how much of this to allocate between two randomly chosen participants.

As an allocatee, you will be randomly matched with another participant twice, with your allocation in each of these matches determined by a different randomly assigned allocator.

Note that as an allocator, the two allocatees you are matched with will never be responsible for allocating for you.

Please click next to proceed.

Next

Allocation Task: Instructions

Allocation Task

As an allocator, suppose you are allocating the 1000 ECUs between two participants.
Please decide on your allocation below.


receives

← 1000 ECUs →


receives

Submit

Allocation Task

Stage 3: Risk Preference Task (All)

A-B Task

Below you will see 10 scenarios.

In each scenario, you will have to make a choice between 2 options, A & B.

After you made your choice, one of the scenarios will be picked randomly based on a 10-sided die roll and your payoff to this task will be based on your choice for that scenario, A or B, and a second 10-sided die roll.

For example, if scenario 6 is picked based on the first die roll, and you picked option A, you will receive 1000 ECUs when the die shows any number from 1 to 6 (probability of 6/10) and 800 ECUs when the die shows any number from 7 to 10 (probability of 4/10) based on a second die roll.

The payoff will be converted to SGD and given to you at the end of this stage.

Since you do not know which scenario will be picked, you will have to pay attention to each of the scenarios and make your decision carefully.

	Option A		Option B
#1:	1000 ECUs with probability 1/10, 800 ECUs with probability 9/10	☐	1900 ECUs with probability 1/10, 100 ECUs with probability 9/10
#2:	1000 ECUs with probability 2/10, 800 ECUs with probability 8/10	☐	1900 ECUs with probability 2/10, 100 ECUs with probability 8/10
#3:	1000 ECUs with probability 3/10, 800 ECUs with probability 7/10	☐	1900 ECUs with probability 3/10, 100 ECUs with probability 7/10
#4:	1000 ECUs with probability 4/10, 800 ECUs with probability 6/10	☐	1900 ECUs with probability 4/10, 100 ECUs with probability 6/10
#5:	1000 ECUs with probability 5/10, 800 ECUs with probability 5/10	☐	1900 ECUs with probability 5/10, 100 ECUs with probability 5/10
#6:	1000 ECUs with probability 6/10, 800 ECUs with probability 4/10	☐	1900 ECUs with probability 6/10, 100 ECUs with probability 4/10
#7:	1000 ECUs with probability 7/10, 800 ECUs with probability 3/10	☐	1900 ECUs with probability 7/10, 100 ECUs with probability 3/10
#8:	1000 ECUs with probability 8/10, 800 ECUs with probability 2/10	☐	1900 ECUs with probability 8/10, 100 ECUs with probability 2/10
#9:	1000 ECUs with probability 9/10, 800 ECUs with probability 1/10	☐	1900 ECUs with probability 9/10, 100 ECUs with probability 1/10
#10:	1000 ECUs with probability 10/10, 800 ECUs with probability 0/10	☐	1900 ECUs with probability 10/10, 100 ECUs with probability 0/10

Submit

Risk Preference Task

F.3 Post-Experiment Survey

General

Was there anything unclear about the instructions? If so, please state so. [Open-ended]

To what extent did you understand the instructions for the experiment?

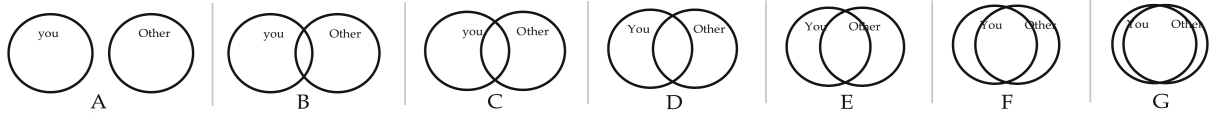
[0: Do not understand at all - 6: Understand completely]

Have you ever participated in an experiment similar to this before? [Yes, No]

§Identity manipulation check

Please select the picture that best describes your feelings towards group members from Group BLUE

Please select the picture that best describes your feelings towards group members from Group RED.



Please indicate the extent to which you agree or disagree with the following statements about group members from your Group BLUE.

I identify with members from my group. [1: Strongly Agree - 5: Strongly Disagree]

I trust members from my group. [1: Strongly Agree - 5: Strongly Disagree]

Members from my group generally trust me. [1: Strongly Agree - 5: Strongly Disagree]

Questions about Decision Task

In Stage 2, how did you decide on which activity you want to join? [Open-ended]

Did you find anything unfair about the experiment? If so, please state so. [Open-ended]

Please answer the following questions about your experience in Stage 2, the Decision Task.

†How cooperative would you rate the other participant(s) from Group BLUE you were matched with in Stage 2? [0: Not cooperative at all - 4: Completely cooperative]

†How would you rate your level of trust for the other participant(s) from Group BLUE you were matched with in Stage 2? [0: Do not trust at all - 4: Trust completely]

†How cooperative would you rate the other participant(s) from Group RED you were matched with in Stage 2? [0: Not cooperative at all - 4: Completely cooperative]

†How would you rate your level of trust for the other participant(s) from Group RED you were matched with in Stage 2? [0: Do not trust at all - 4: Trust completely]

Demographic questions

Year of study [Year 1, Year 2, Year 3, Year 4, Year 5, Postgraduate, Not Applicable]

Gender [Female, Male, Prefer not to say]

Course of study [Open-ended]

Nationality [Open-ended]

Ethnicity [Open-ended]

§ Not included for no identity treatment. Identity questions apply to blue members, replace BLUE with RED and vice versa for red members. Items were randomized within section.

† In the no identity treatment, there was no mention of BLUE or RED. In homogeneous groups, either the BLUE or RED questions were asked. In heterogeneous groups, all questions were asked.